
Software Requirements Specification For QuitMate Tobacco Awareness

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1. Introduction

QuitMate - Tobacco Awareness is a mobile application designed to help people quit smoking and other tobacco products. The application supports tobacco quitters by providing step-by-step guidance, daily goals, craving management tools, emotional support, and motivational features.

Many people want to quit tobacco but face difficulties because of addiction, stress, social influence, or lack of support. This system simplifies the quitting journey by dividing the process into manageable daily activities and providing assistance during difficult moments.

The application also helps tobacco quitters monitor their progress, track money saved from avoiding tobacco products, and set personal savings goals for meaningful rewards.

Overall, QuitMate - Tobacco Awareness aims to support tobacco quitters in achieving a healthier and tobacco-free lifestyle.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a clear and structured description of the QuitMate - Tobacco Awareness mobile-based tobacco cessation system.

This document describes the system objectives, features, functional requirements, non-functional requirements, and limitations to ensure a proper understanding among all stakeholders, including tobacco quitters, peer group members, moderators, medical advisors, system administrators, and project authorities.

The document serves as a guideline for system analysis, design, development, testing, and maintenance. It also defines the expected behavior and performance of the system and acts as a reference for the development team, supervisors, clients, and other stakeholders involved in the project.

1.2 Project Scope

QuitMate - Tobacco Awareness is a mobile application that helps tobacco quitters stop tobacco use through a structured and supportive quitting process.

The system allows tobacco quitters to:

1. Create a profile with basic information such as age, gender, institution, and tobacco type.
2. Select a personal quit date.
3. Receive a personalized quit plan for 7, 14, or 30 days.
4. Set daily quitting goals.
5. Identify personal tobacco triggers such as stress, friends, or after meals.

-
6. Complete daily check-ins regarding tobacco usage.
 7. Track craving levels using a scale from 1 to 10.
 8. Monitor mood conditions such as stress, anxiety, and boredom.
 9. Write short daily reflections and progress notes.
 10. Use an SOS button during strong cravings.
 11. Follow a 5-minute survival timer during cravings.
 12. Practice guided breathing exercises using the 4-7-8 breathing method.
 13. Access quick distraction techniques during cravings.
 14. Track money saved by avoiding tobacco products.
 15. Create a wishlist goal using saved money.
 16. View savings progress through a progress bar.
 17. Join peer support group chats.
 18. Send messages to peer group members.
 19. Ask anonymous questions.
 20. Request immediate support when necessary.
 21. Earn badges for completing activities.
 22. Increase levels such as Beginner, Warrior, and Smoke-Free Champion.
 23. Read tobacco awareness articles and educational content.
 24. Receive motivational notifications and reminders.

The system mainly focuses on behavior change, emotional support, motivation, awareness, and progress tracking.

2. Stakeholders

The system contains different categories of stakeholders. Each stakeholder has specific roles, responsibilities, and system requirements.

2.1 Primary Stakeholders

2.1.1 Tobacco Quitter

Description

The primary stakeholder of the system who uses the application to quit tobacco products.

Needs

1. Easy and personalized quit plans
2. Daily motivational support
3. Craving management assistance
4. Progress monitoring tools
5. Financial savings tracking
6. Peer and peer member support
7. Awareness and educational materials

Responsibilities

1. Provide accurate information during registration
2. Use the application honestly and responsibly
3. Follow community guidelines and rules

2.1.2 Peer Group Member**Description**

A support person who provides emotional guidance and encouragement to tobacco quitters.

Needs

1. Access to support requests
2. Secure messaging system
3. Communication tools for guidance and peer support

Responsibilities

1. Respect tobacco quitter privacy
2. Provide supportive and appropriate advice
3. Follow platform policies and ethical standards

2.1.3 Moderator**Description**

A stakeholder responsible for monitoring community discussions and maintaining a safe communication environment.

Needs

-
1. Access to group chats and reports
 2. Ability to remove inappropriate content
 3. Monitoring and moderation tools

Responsibilities

1. Enforce community guidelines
2. Maintain respectful discussions
3. Prevent misuse of the platform

2.1.4 Medical Advisor**Description**

A healthcare or awareness professional who provides verified quitting guidelines, awareness articles, and day-wise quitting tasks for the system. The development team collects and manages this information from medical advisors for use inside the application.

Needs

1. Ensure awareness information is medically accurate
2. Provide verified quitting guidance and recommendations
3. Support the preparation of educational content

Responsibilities

1. Provide authentic and evidence-based information
2. Review awareness materials and quitting guidance
3. Ensure the accuracy and reliability of health-related content

2.1.5 System Administrator**Description**

The stakeholder responsible for managing and maintaining the entire system.

Needs

1. Manage tobacco quitters, community members, and moderators
2. Manage awareness content and system settings
3. Monitor system performance and security

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4. Handle backups and technical maintenance

Responsibilities

1. Maintain system stability and security
2. Perform system updates and backups
3. Resolve technical and operational issues

2.2 Secondary Stakeholders

2.2.1 Project Owner (Authority)

Description

The authority responsible for overseeing and approving the project.

Needs

1. Monitor system effectiveness and public impact
2. Approve major updates and improvements
3. Ensure achievement of public health objectives

Responsibilities

1. Provide overall project direction
2. Approve major decisions and policies
3. Support long-term system sustainability

3. Requirement Specification

All the requirements based on the elicitation process are described in this section.

3.1 Business Requirements

BR-1: Provide Structured Tobacco Cessation Support System

Field	Content	Field	Content
BR-1	The system shall provide a structured mobile-based tobacco cessation support system for tobacco quitters.		
Description	The system shall offer personalized quit plans, craving management tools, emotional support features, savings		

	tracking, awareness materials, and motivational mechanisms to assist tobacco quitters in achieving a tobacco-free lifestyle.		
Stakeholders	Tobacco Quitter, Peer Group Member, Medical Advisor, System Administrator, Project Owner	Priority	High

BR-2: Enable Digital Peer Support and Peer Support Services

Field	Content	Field	Content
BR-2	The system shall enable secure peer support and digital peer support services.		
Description	The system shall support peer group discussions, peer support messaging, anonymous questions, and emergency support requests under moderation and administrative supervision.		
Stakeholders	Tobacco Quitter, Peer Group Member, Moderator, System Administrator	Priority	High

3.2 Business Objectives

BO-1: Improve tobacco quitting success rate

Field	Content	Field	Content
BO-1	The system shall improve quitting success rates through structured behavioral guidance and monitoring.		
Description	Tobacco Awareness shall support daily check-ins, craving tracking, survival tools, emotional monitoring, savings visualization, and motivational rewards to encourage sustained quitting behavior.		
Stakeholders	Tobacco Quitter, Medical Advisor, Project Owner	Priority	High

BO-2: Promote Public Health Awareness and Behavioral Change

Field	Content	Field	Content
BO-2	The system shall promote awareness of tobacco-related health risks and encourage healthy lifestyle habits.		

Description	The system shall provide verified educational content, awareness articles, day-wise quitting guidance, and motivational notifications to increase knowledge and prevent relapse.		
Stakeholders	Tobacco Quitter, Medical Advisor, Project Owner	Priority	High

3.3 User Requirements

3.3.1 Primary Users

UR-P1: Personalized Quit Planning

Field	Content	Field	Content
UR-P1	The system shall allow tobacco quitters to create a personalized quit plan.		
Description	Tobacco quitters shall select a quit date and receive a structured 7, 14, or 30-day quitting plan with daily goals.		
Stakeholders	Tobacco Quitter, Medical Advisor	Priority	High

UR-P2: Daily Check-in and Craving Tracking

Field	Content	Field	Content
UR-P2	The system shall allow tobacco quitters to complete daily check-ins and track cravings.		
Description	Tobacco quitters shall record daily tobacco usage status, craving level (1–10 scale), mood condition, and short reflections.		
Stakeholders	Tobacco Quitter	Priority	High

UR-P3: Craving Emergency Support Tools

Field	Content	Field	Content
UR-P3	The system shall provide emergency craving support tools.		

Description	Tobacco quitters shall access SOS support, survival timer, breathing exercises (4-7-8 method), and quick distraction activities during cravings.		
Stakeholders	Tobacco Quitter	Priority	High

UR-P4: Savings and Reward Tracking

Field	Content	Field	Content
UR-P4	The system shall allow tobacco quitters to track money saved from quitting tobacco.		
Description	The system shall calculate daily, weekly, and monthly savings and allow tobacco quitters to create wishlist goals with progress visualization.		
Stakeholders	Tobacco Quitter	Priority	High

UR-P5: Peer Support and Peer Support Communication

Field	Content	Field	Content
UR-P5	The system shall allow tobacco quitters to communicate with peer group members and support groups.		
Description	Tobacco quitters shall join peer support groups, send messages to community members, request support, and ask anonymous questions.		
Stakeholders	Tobacco Quitter, Peer Group Member	Priority	High

3.3.2 Secondary Users

UR-S1: Peer Support Message Delivery

Field	Content	Field	Content
UR-S1	The system shall allow peer group members to send motivational and peer support messages.		
Description	Peer community members shall provide emotional guidance and respond to support requests securely.		
Stakeholders	Peer Group Member, Tobacco Quitter	Priority	High

UR-S2: Administrative System Management

Field	Content	Field	Content
UR-S2	The system shall allow administrators to manage users, community members, and content.		
Description	Administrators shall manage awareness materials, peer member accounts, reports, system performance, and moderation controls.		
Stakeholders	System Administrator, Moderator	Priority	High

3.4 Business Rules

BRule-1: Verified Awareness Content Only

Field	Content	Field	Content
BRule-1	The system shall publish awareness materials only after administrative verification.		
Description	Educational content provided by medical advisors shall be reviewed and approved before being visible to tobacco quitters.		
Stakeholders	Medical Advisor, System Administrator	Priority	High

BRule-2: Secure and Ethical Communication

Field	Content	Field	Content
BRule-2	The system shall enforce respectful and secure communication guidelines.		
Description	Moderators shall monitor peer discussions and peer support communications to prevent misuse or inappropriate content.		
Stakeholders	Moderator, Peer Group Member, Tobacco Quitter	Priority	High

3.5 Functional Requirements

FR1 – User Registration & Profile Management: The system shall provide secure user registration, authentication, and baseline profile assessment.

Field	Content	Field	Content
FR-1	The system shall allow tobacco quitters to manage profile information and set a quit date.		
Description	Tobacco quitters shall update personal details and select a target quit date which will be stored and used for quit plan generation.		
Stakeholders	Tobacco Quitter, System Administrator	Priority	High

FR2 – Quit Date & Financial Baseline Setup: The system shall allow quitters to schedule a target quit date, specify pack costs, and calculate daily baseline expenditure.

Field	Content	Field	Content
FR-2	The system shall generate and display personalized quit plans.		
Description	The system shall provide structured 7, 14, or 30-day quit plans with daily goals and milestone tracking.		
Stakeholders	Tobacco Quitter, Medical Advisor	Priority	High

FR3 – Daily Check-in & Habit Tracking: The system shall log daily cravings intensity (1-10), mood rating, and cigarettes smoked count while updating clean streaks and awarding badges.

Field	Content	Field	Content
FR-3	The system shall allow tobacco quitters to complete daily check-ins.		
Description	The system shall record daily tobacco usage status, craving levels (1–10), mood conditions, and reflections for progress tracking.		
Stakeholders	Tobacco Quitter	Priority	High

FR4 – Financial Savings Calculation & Wishlist Tracking: The system shall calculate total financial savings accumulated since quit date and track percentage progress toward wishlist goals.

Field	Content	Field	Content
FR-4	The system shall calculate and display money saved from avoiding tobacco use.		

Description	The system shall calculate daily, weekly, and monthly savings and allow tobacco quitters to set wishlist goals with progress tracking.		
Stakeholders	Tobacco Quitter	Priority	High

FR5 – Emergency SOS Craving Support: The system shall provide an emergency SOS hub featuring a 5-Minute Survival Timer, Guided 4-7-8 Breathing exercises, and Quick Distraction tools.

Field	Content	Field	Content
FR-5	The system shall provide emergency craving support tools.		
Description	The system shall provide SOS support, survival timer, breathing exercise (4-7-8), and quick distraction activities during strong cravings.		
Stakeholders	Tobacco Quitter	Priority	High

FR6 – Peer Support Community Chat: The system shall provide a real-time community chatroom for tobacco quitters to share encouragement and peer motivation.

Field	Content	Field	Content
FR-6	The system shall support peer group communication and peer support messaging.		
Description	The system shall allow tobacco quitters to join support groups, send messages to peer group members, and request support.		
Stakeholders	Tobacco Quitter, Peer Group Member, Moderator	Priority	High

FR7 – Groq AI Craving Assistant Integration: The system shall integrate with Groq AI Service API to deliver intelligent, personalized craving coping advice.

Field	Content	Field	Content
FR-7	The system shall implement gamification features to motivate tobacco quitters.		
Description	The system shall assign badges, levels (Beginner, Warrior, Champion), streaks, and badges based on completed activities.		

Stakeholders	Tobacco Quitter, System Administrator	Priority	Medium
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FR8 – Tobacco Awareness Articles & Health Timeline: The system shall present educational awareness articles and a health recovery timeline (20m to 1yr recovery milestones).

Field	Content	Field	Content
FR-8	The system shall send reminders and motivational notifications.		
Description	The system shall generate scheduled and event-based notifications for daily check-ins, milestones, and awareness content.		
Stakeholders	Tobacco Quitter, System Administrator	Priority	High

FR9 – Offline Data Caching & Auto-Synchronization: The system shall locally cache user entries in Hive offline storage, auto-syncing with the PostgreSQL backend when internet connectivity is restored.

Field	Content	Field	Content
FR-9	The system shall locally cache user check-ins, profile edits, and savings goals in Hive offline storage, auto-syncing with PostgreSQL database when internet connectivity is restored.		
Description	The system shall automatically synchronize pending offline Hive caches with the PostgreSQL backend database upon network restoration.		
Stakeholders	Tobacco Quitter, System Administrator, PostgreSQL Database	Priority	Medium

3.6 Quality Attributes / Performance Requirements

QA-1: System Availability

Field	Content	Field	Content
QA-1	The system shall maintain 99% monthly availability for all core features.		

Description	The system shall remain operational with high uptime so tobacco quitters can access support tools, check-ins, and resources at any time.		
Stakeholders	Tobacco Quitter, System Administrator, Project Owner	Priority	High

QA-2: Response Time

Field	Content	Field	Content
QA-2	Standard operations shall respond within 2 seconds.		
Description	The system shall provide fast response for profile updates, daily check-ins, craving tools, savings calculation, and content loading.		
Stakeholders	Tobacco Quitter	Priority	High

QA-3: Offline Data Synchronization Speed

Field	Content	Field	Content
QA-3	Offline data synchronization shall complete within 3 seconds under normal network conditions.		
Description	The system shall ensure quick and reliable local caching of user check-in entries during offline operation.		
Stakeholders	Tobacco Quitter, System Administrator, Project Owner	Priority	Medium

QA-4: Security and Privacy

Field	Content	Field	Content
QA-4	All user data shall be securely encrypted and role-protected.		
Description	The system shall protect personal information, health-related data, messages, and progress records using encryption and proper access control.		
Stakeholders	Tobacco Quitter, System Administrator, Project Owner	Priority	High

QA-5: Scalability

Field	Content	Field	Content
QA-5	The system shall scale to support increasing users and data growth.		
Description	The system shall maintain stable performance as the number of tobacco quitters, messages, and daily records increases.		
Stakeholders	Project Owner, System Administrator	Priority	Medium

3.7 External Interface Requirements

EIR-1: PostgreSQL Database Interface

Field	Content	Field	Content
EIR-1	The system shall integrate with external offline cache services.		
Description	The system shall connect with secure financial savings services to process entries for app features and peer support sessions.		
Stakeholders	Tobacco Quitter, System Administrator, Project Owner	Priority	High

EIR-2: Notification Service Interface

Field	Content	Field	Content
EIR-2	The system shall integrate with notification services (SMS/Push).		
Description	The system shall connect with external services to deliver motivational reminders, craving alerts, and daily check-in notifications.		
Stakeholders	Tobacco Quitter, System Administrator	Priority	High

EIR-3: Database Service Interface

Field	Content	Field	Content
EIR-3	The system shall interface with secure database services.		

Description	The system shall securely store and retrieve all user profiles, quit plans, daily records, messages, and savings data.		
Stakeholders	System Administrator, Project Owner	Priority	High

3.8 Design Constraints

DC-1: Mobile-First Design

Field	Content	Field	Content
DC-1	The system shall follow a mobile-first design approach.		
Description	The application shall be designed primarily for smartphones with simple and easy navigation suitable for tobacco quitters.		
Stakeholders	Tobacco Quitter, Project Owner	Priority	High

DC-2: Secure Communication

Field	Content	Field	Content
DC-2	The system shall support secure HTTPS communication.		
Description	All data transmission between the mobile application and server shall use secure HTTPS protocol.		
Stakeholders	System Administrator, Project Owner	Priority	High

DC-3: Public Health Data Protection

Field	Content	Field	Content
DC-3	The system shall follow public health data protection standards.		
Description	The system shall comply with health data privacy guidelines while handling sensitive information related to tobacco quitting and emotional state.		
Stakeholders	Medical Advisor, System Administrator, Project Owner	Priority	High

3.9 Data Requirements

DR-1: User Profile and Quit Plan Data

Field	Content	Field	Content
DR-1	The system shall store user profile and quit plan data.		
Description	Data shall include age, gender, tobacco type, quit date, personalized quit plan, and daily goals.		
Stakeholders	Tobacco Quitter, System Administrator	Priority	High

DR-2: Daily Check-in and Craving Records

Field	Content	Field	Content
DR-2	The system shall store daily check-in and craving records.		
Description	Data shall include daily usage status, craving level (1-10), mood conditions, reflections, and survival timer usage.		
Stakeholders	Tobacco Quitter, Medical Advisor	Priority	High

DR-3: Messaging and Peer Support Records

Field	Content	Field	Content
DR-3	The system shall store messaging and peer support records securely.		
Description	Data shall include peer group messages, peer support conversations, support requests, and anonymous questions.		
Stakeholders	Tobacco Quitter, Peer Group Member, System Administrator	Priority	High

DR-4: Financial Savings and Wishlist Goal Records

Field	Content	Field	Content
DR-4	The system shall store financial savings goal records and wishlist milestone progress.		
Description	Data shall include money saved, wishlist goals, progress bar status, and financial savings history for app features.		
Stakeholders	Tobacco Quitter, System Administrator	Priority	Medium

DR-5: Notification and System Logs

Field	Content	Field	Content
DR-5	The system shall store notification logs and delivery history.		
Description	Data shall include sent notifications, motivational messages, delivery status, and system activity logs.		
Stakeholders	System Administrator	Priority	Medium

4. Requirement Engineering Process

The Requirement Engineering process for the QuitMate - Tobacco Awareness mobile application involved systematic identification, analysis, documentation, and validation of requirements from all relevant stakeholders. The primary goal was to understand the behavioral, emotional, technical, and health-related needs of tobacco quitters and support providers. The process ensured that the system requirements are accurate, complete, feasible, and aligned with public health objectives.

Multiple elicitation techniques were used to gather comprehensive information from stakeholders including tobacco quitters, peer group members, medical advisors, moderators, the system administrator, and the project authority.

4.1 Requirement Elicitation Techniques

Requirement elicitation for QuitMate - Tobacco Awareness used several complementary techniques to ensure a clear understanding of user expectations, quitting challenges, emotional triggers, and system usability requirements. Each technique targeted specific stakeholder groups to ensure balanced and reliable information gathering.

4.1.1 Interviews

Purpose:

The purpose of interviews was to collect detailed qualitative insights about tobacco quitting challenges, craving behavior, emotional triggers, and expectations from a digital cessation support system.

Approach:

1. Semi-structured one-on-one interviews were conducted.
2. Interviews were held both face-to-face and online.
3. Each interview lasted approximately 20 to 30 minutes.

Stakeholders Involved:

1. Ten tobacco quitters from different age groups.

-
2. Two peer group members.
 3. One medical advisor.

Example Interview Questions:

1. What are the main difficulties you face when trying to quit tobacco?
2. What situations usually trigger your cravings?
3. Would daily reminders help you stay committed to quitting?
4. How important is emotional support during strong cravings?
5. Would tracking money saved motivate you to continue quitting?
6. Which support tool do you prefer during cravings: breathing exercises or distraction activities?

Findings:

1. Most tobacco quitters struggle due to stress, social influence, and habitual triggers.
2. Strong cravings typically last between 3 to 10 minutes.
3. Immediate support tools increase confidence and control.
4. Financial savings tracking improves motivation.
5. Emotional encouragement significantly improves quitting consistency.

Impact on SRS:

The interview findings directly influenced the inclusion of features such as the survival timer, SOS support button, breathing exercise module, savings tracker, daily check-in system, and peer peer support communication.

4.1.2 Workshops

Purpose:

The purpose of workshops was to collaboratively identify system priorities, quitting flow, feature importance, and interaction design among stakeholders.

Method:

1. A workshop was conducted in a university health awareness session.
2. Participants included five tobacco quitters, two peer group members, and one faculty advisor.
3. Brainstorming and feature prioritization techniques were applied.

Outputs:

1. The quitting journey flow was defined as Profile Setup, Quit Date Selection, Personalized Plan, Daily Check-in, Craving Support, and Progress Tracking.
2. The need for an emergency SOS feature was confirmed.
3. A five-minute survival timer was strongly recommended.
4. Gamification elements such as Streak Badges and levels were suggested.
5. Wishlist goals based on saved money were proposed.

Result:

Workshop outcomes significantly shaped the gamification requirements, craving management tools, and milestone visualization features.

4.1.3 Focus Group Discussion**Purpose:**

The purpose of the focus group discussion was to explore emotional, behavioral, and motivational aspects of quitting tobacco collectively.

Setup:

1. Eight participants were involved, including six tobacco quitters and two peer group members.
2. The session lasted approximately one hour and was moderated.

Key Discussion Topics:

1. Emotional triggers and relapse risks.
2. Importance of peer support.
3. Role of daily motivational messages.
4. Clarity and simplicity of awareness content.

Findings:

1. Many quitters feel isolated during relapse moments.
2. Group support reduces relapse risk.
3. Short motivational messages are highly effective.
4. Awareness content must be simple and easy to understand.

Impact:

The discussion resulted in the inclusion of peer support group chat, peer support messaging, motivational notifications, and structured awareness content sections.

4.1.4 Observation

Purpose:

The purpose of observation was to understand real behavioral patterns during cravings and stress situations.

Process:

1. Quitting attempts were observed during awareness sessions.
2. Reactions during simulated craving situations were monitored.
3. Stress-handling behavior and coping strategies were recorded.

Insights:

1. Cravings usually peak within a short duration of around five minutes.
2. Guided breathing reduces anxiety levels.
3. Distraction activities reduce urge intensity.

Result:

Observation findings inspired the implementation of the five-minute survival timer, guided breathing animation, and quick distraction module.

4.1.5 Questionnaires

Purpose:

The purpose of questionnaires was to collect quantitative data to validate system requirements on a larger scale.

Respondents:

1. Twenty tobacco quitters.
2. Five peer group members.

Sample Questions for Tobacco Quitters:

1. What usually triggers your tobacco cravings?
2. Would a daily progress tracker help you stay motivated?
3. Do you prefer group support or private peer support?
4. Would tracking saved money encourage you to continue quitting?

-
5. How useful would breathing exercises be during cravings?
 6. Would gamification features such as Streak Badges and levels increase your engagement?

Sample Questions for Peer Group Members:

1. What type of support do tobacco quitters need most?
2. Should peer support messages be instant or scheduled?
3. How important is monitoring craving levels?
4. Do you require tools to monitor user progress?
5. Should the system allow anonymous questions?

Findings:

1. Ninety percent of respondents prefer daily reminders.
2. Eighty-five percent want craving tracking tools.
3. Seventy-eight percent believe peer support improves quitting success.
4. Seventy-two percent prefer money-saving visualization.
5. Eighty-eight percent support gamification features.

Impact:

The questionnaire results confirmed the importance of daily check-ins, craving scales, savings progress bars, peer communication features, and Streak Badges-based motivational systems.

4.1.6 User Interface Analysis

Purpose:

The purpose of user interface analysis was to evaluate usability and accessibility expectations for mobile users.

Process:

1. Low-fidelity mobile wireframes were created.
2. Navigation patterns were observed.
3. User feedback on layout and design was collected.

Findings:

1. Users prefer calm and simple color schemes.
2. Large buttons are necessary for the SOS feature.

-
3. A simple dashboard is preferred over complex graphs.
 4. Quick access to craving support tools is essential.

Impact:

These findings influenced the mobile-first design constraint, simplified dashboard structure, large SOS button placement, and clear progress indicators.

4.2 Sample of Requirement Collection

This section demonstrates how raw stakeholder inputs were transformed into formal system requirements.

TQ-01: Tobacco Quitter want immediate help during cravings.

The system shall provide a survival timer and breathing exercise module.

TQ-02: Tobacco Quitter want financial motivation.

The system shall calculate and display money saved from avoiding tobacco products.

PC-03: Community Members require communication tools.

The system shall support secure peer support messaging and support requests.

MED-04: Verified awareness content is required.

The system shall publish awareness materials only after proper verification.

PERF-05: Users expect fast system response.

The system shall respond within two seconds under normal operating conditions.

All documented requirements are traceable to their elicitation sources.

4.3 Requirement Validation

Requirement validation ensures that the documented requirements for the QuitMate - Tobacco Awareness system are correct, complete, consistent, and feasible. Even when requirements appear clear during documentation, ambiguities or conflicts may arise during development. Therefore, comprehensive validation is essential.

The validation process ensures that the system effectively supports tobacco quitting through structured plans, craving management tools, emotional support features, savings tracking, and awareness content in a reliable and user-friendly manner.

4.3.1 Review the Requirements

A structured review process was conducted involving the development team, academic supervisor, peer group members, and the medical advisor.

Review activities focused on:

1. Functional and non-functional requirements.

-
2. User workflows for tobacco quitters and community members.
 3. Emotional safety and ethical communication.
 4. Privacy and data protection.
 5. Feasibility of mobile implementation.

This process helped identify ambiguities, inconsistencies, and unrealistic expectations early, reducing potential development risks.

4.3.2 Test the Requirements

Requirement-based validation tests were defined to ensure:

1. Each requirement is measurable and verifiable.
2. Craving support tools function correctly.
3. Messaging services operate reliably.
4. Savings calculations are accurate.
5. Notifications are triggered appropriately.

Testing requirements before full implementation ensures that development aligns with defined expectations.

4.3.3 Simulate the Requirements

Simulation-based validation was conducted using interactive prototypes to visualize system behavior before implementation.

Simulation activities included:

1. Clickable mobile interface mockups.
2. Craving support workflow demonstrations.
3. Quit plan generation simulations.
4. Messaging and notification scenarios.
5. Savings tracking and gamification flow demonstrations.

Simulation enabled early detection of usability issues and workflow gaps, improving overall system quality prior to development.

5. Use Case Diagram

Figure 1: Use Case Diagram for QuitMate - Tobacco Awareness

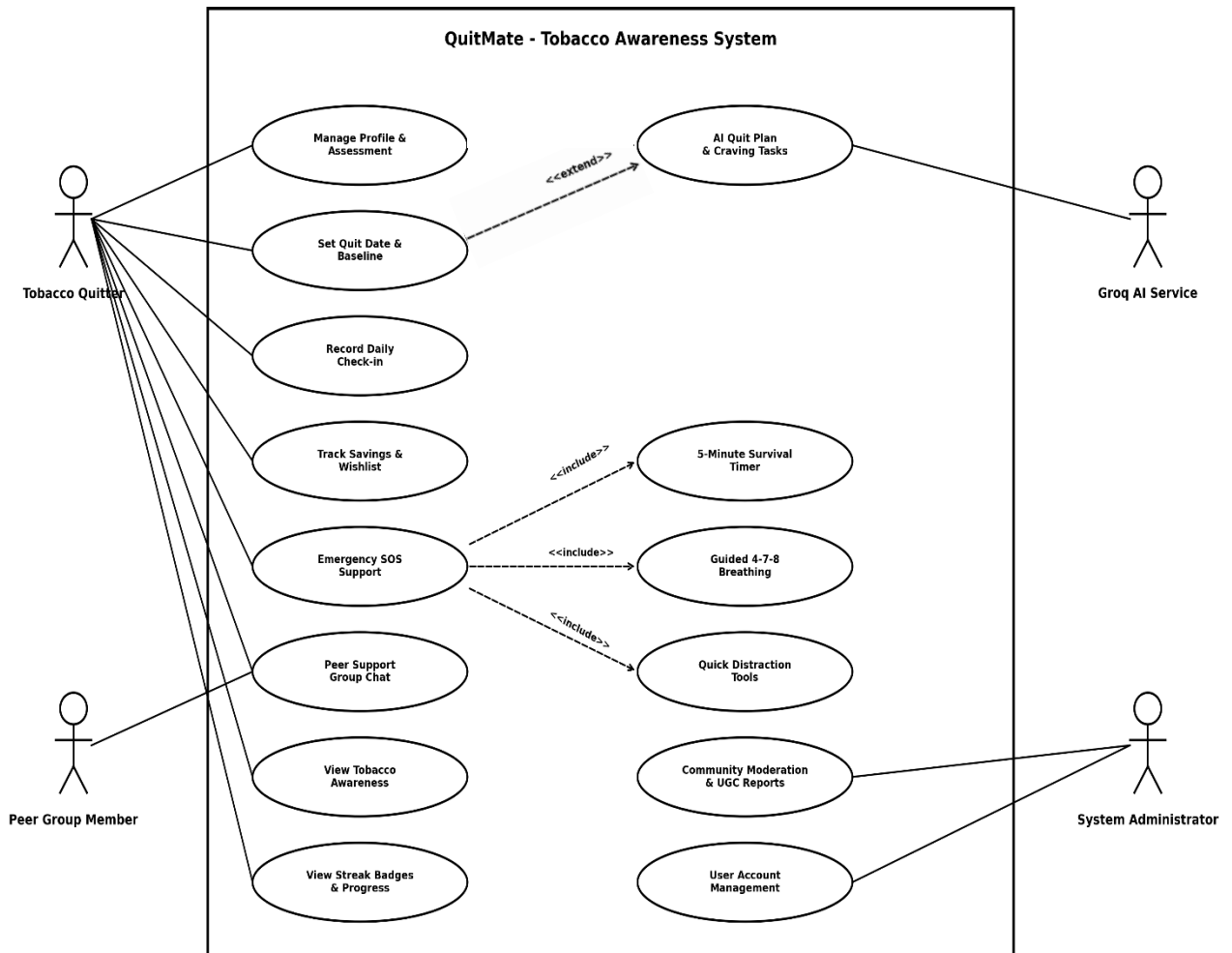


Figure 1: Use Case Diagram for QuitMate - Tobacco Awareness

6. Use case Description

6.1 UC-01 : Manage Profile

Field	Description
Use Case No	UC-01
Use Case Name	Manage Profile & Assessment
Goal	To allow the tobacco quitter to create, view, and update personal profile details and baseline smoking assessment.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Profile and initial assessment details are successfully recorded in the system.
Failed End Condition	Profile update fails due to invalid form input.
Trigger	The tobacco quitter opens the profile assessment section.
Main Success Flow	1. The tobacco quitter opens the profile assessment screen. 2. The system presents profile fields and smoking habit assessment questions. 3. The tobacco quitter enters personal details, cigarettes smoked per day, and stick price. 4. The system validates the entered information. 5. The system stores the updated profile and calculates initial baseline stats. 6. The system displays a profile updated confirmation.
Alternative Flows	A1: Invalid Profile Input 4a. The system detects missing or invalid profile data. 4b. The system displays specific field error messages. 4c. The tobacco quitter corrects the data and resubmits.
Quality Attributes	1. Security: Personal profile data shall be protected. 2. Usability: Form fields shall be intuitive and clear. 3. Reliability: User profile data shall be stored accurately.

6.2 UC-02 : Set Quit Date

Field	Description
Use Case No	UC-02

Use Case Name	Set Quit Date & Baseline
Goal	To allow the tobacco quitter to select a target quit date and establish a baseline smoking cost.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	The target quit date and financial baseline parameters are saved.
Failed End Condition	Quit date setting fails due to invalid date selection.
Trigger	The tobacco quitter selects the Set Quit Date feature.
Main Success Flow	1. The tobacco quitter opens the Quit Plan setup section. 2. The system prompts for a target quit date and cigarette cost. 3. The tobacco quitter selects the target date and cost per stick/pack. 4. The system validates the date parameter. 5. The system saves the target quit date and calculates daily baseline cost. 6. The system confirms successful setup.
Alternative Flows	A1: Past Date Selected 4a. The system detects a selected date prior to allowed range. 4b. The system prompts the quitter to select a valid future date. 4c. The tobacco quitter selects a valid date.
Quality Attributes	1. Usability: Date picker shall provide easy selection. 2. Accuracy: Daily cost baseline calculations shall be precise.

6.3 UC-03 : View Quit Plan

Field	Description
Use Case No	UC-03
Use Case Name	View & Personalize Quit Plan
Goal	To allow the tobacco quitter to view milestone stages, health recovery timelines, and personalized quit plan recommendations.
Primary Actors	Tobacco Quitter

Secondary Actors	None
Precondition	N/A
Success End Condition	The quit plan milestones and recovery progress are displayed.
Failed End Condition	Unable to load quit plan details.
Trigger	The tobacco quitter navigates to the Quit Plan screen.
Main Success Flow	1. The tobacco quitter opens the Quit Plan section. 2. The system retrieves stage milestones and progress calculations based on quit date. 3. The system renders stage cards, recovery timeline, and personalized advice. 4. The tobacco quitter views plan stages.
Alternative Flows	A1: Network Disconnection 2a. The system detects lack of server connectivity. 2b. The system retrieves locally cached quit plan data. 2c. The quit plan is displayed in offline mode.
Quality Attributes	1. Performance: Quit plan screen shall load within 1.5 seconds. 2. Usability: Stages shall be presented visually.

6.4 UC-04 : Daily Check-in

Field	Description
Use Case No	UC-04
Use Case Name	Daily Check-in
Goal	To log daily smoking status, craving intensity, mood rating, and update clean streak progress.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Daily check-in entry is recorded, clean streak is incremented, and badges are awarded.
Failed End Condition	Check-in submission fails due to incomplete form data.

Trigger	The tobacco quitter initiates daily check-in from dashboard.
Main Success Flow	1. The tobacco quitter opens the Daily Check-in screen. 2. The system prompts for craving intensity score, mood rating, and cigarettes smoked count. 3. The tobacco quitter completes the entries and submits. 4. The system validates the check-in data. 5. The system records the check-in, updates clean streak, and computes earned Streak Badges. 6. The system displays check-in summary and streak animation.
Alternative Flows	A1: Duplicate Check-in Today 4a. The system detects a check-in already logged for today. 4b. The system allows updating existing check-in entries. 4c. The updated entry is saved.
Quality Attributes	1. Usability: Check-in form shall take under 30 seconds to complete. 2. Reliability: Streak calculations shall be strictly accurate.

6.5 UC-05 : Track Saving

Field	Description
Use Case No	UC-05
Use Case Name	Track Financial Savings & Wishlist
Goal	To calculate financial money saved since quit date and track progress toward user wishlist savings goals.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Total money saved is calculated and wishlist goal progress is updated.
Failed End Condition	Wishlist goal creation fails due to invalid price input.
Trigger	The tobacco quitter opens the Money Saver screen.
Main Success Flow	1. The tobacco quitter navigates to Money Saver screen. 2. The system calculates accumulated money saved based on quit date and daily cost. 3. The tobacco quitter views total saved amount and existing wishlist items. 4. The tobacco quitter adds a new wishlist item with target cost. 5. The system

	computes progress percentage for each goal. 6. The system displays active savings progress.
Alternative Flows	A1: Invalid Target Price 5a. The system detects zero or negative target price. 5b. The system prompts for a valid positive amount. 5c. The quitter corrects the amount.
Quality Attributes	1. Accuracy: Financial math calculations shall be exact. 2. Usability: Visual progress bars shall represent percentage completed.

6.6 UC-06 : Craving Support

Field	Description
Use Case No	UC-06
Use Case Name	Trigger Emergency SOS Craving Support
Goal	To provide immediate, high-priority craving relief options when a tobacco quitter encounters a sudden smoking urge.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	SOS emergency support hub is presented with immediate intervention options.
Failed End Condition	SOS tool launch fails due to interface error.
Trigger	The tobacco quitter taps the red emergency SOS button.
Main Success Flow	1. The tobacco quitter taps the emergency SOS button on dashboard. 2. The system launches the high-priority SOS Support Hub. 3. The system presents quick craving options: 5-Minute Timer, Guided 4-7-8 Breathing, and Distraction Tools. 4. The system logs the SOS event for craving analytics. 5. The quitter selects an emergency intervention mode.
Alternative Flows	A1: Quick Exit 5a. The quitter closes the SOS hub. 5b. The system logs short SOS duration. 5c. The dashboard returns to normal state.
Quality Attributes	1. Availability: SOS button shall be instantly accessible from main screen. 2. Performance: SOS hub shall open in under 0.5 seconds.

6.7 UC-07 : Start Survival Timer

Field	Description
Use Case No	UC-07
Use Case Name	Start 5-Minute Survival Timer
Goal	To initiate a 300-second countdown timer to help the quitter endure and overcome a peak craving impulse.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	The 5-minute survival timer completes, craving urge subsides, and completion feedback is shown.
Failed End Condition	Timer is cancelled prematurely by user.
Trigger	The quitter selects 5-Minute Survival Timer from SOS hub.
Main Success Flow	1. The tobacco quitter selects 5-Minute Survival Timer. 2. The system initializes a 300-second countdown clock with motivational quotes. 3. The timer ticks down second-by-second while displaying survival tips. 4. The countdown reaches 0:00. 5. The system plays a completion audio signal and displays a success badge. 6. The quitter confirms feeling relieved.
Alternative Flows	A1: User Cancels Timer Early 3a. The quitter taps cancel button before 300 seconds elapse. 3b. The system stops countdown and offers breathing tool instead. 3c. The quitter switches tool.
Quality Attributes	1. Usability: Countdown text shall be large and highly visible. 2. Reliability: Timer tick loop shall run smoothly in foreground.

6.8 UC-08 : Breathing Exercise

Field	Description
Use Case No	UC-08

Use Case Name	Perform Guided 4-7-8 Breathing Exercise
Goal	To guide the tobacco quitter through calming 4-7-8 deep breathing cycles (Inhale 4s, Hold 7s, Exhale 8s) to reduce craving anxiety.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Breathing cycles complete and user experiences reduced physical tension.
Failed End Condition	Breathing session interrupted by user.
Trigger	The quitter selects Guided Breathing from SOS hub.
Main Success Flow	1. The tobacco quitter selects Guided Breathing exercise. 2. The system displays an animated breathing circle with audio cues. 3. The system leads user through 4s Inhale -> 7s Hold -> 8s Exhale phases. 4. The cycle repeats for designated duration. 5. The system finishes the session with calming audio feedback.
Alternative Flows	A1: Audio Disabled 2a. Device audio is muted. 2b. The system continues visual color-coded animation cues. 2c. Session completes visually.
Quality Attributes	1. Usability: Visual animations shall expand and contract rhythmically. 2. Performance: Smooth 60fps graphic transitions.

6.9 UC-09 : Quick Distraction

Field	Description
Use Case No	UC-09
Use Case Name	Quick Distraction Activities
Goal	To engage the tobacco quitter in quick physical or sensory activities (drinking water, short walk, sound effect) to divert attention from cravings.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A

Success End Condition	Distraction activity is completed and logged.
Failed End Condition	Distraction activity is dismissed.
Trigger	The quitter selects Quick Distractions from SOS hub.
Main Success Flow	1. The quitter opens Quick Distractions. 2. The system presents distraction choices: Drink Water Tracker, Walk Counter, Sound Distraction. 3. The quitter selects an activity (e.g., Drink Water). 4. The system executes activity actions and plays interactive sound. 5. The quitter confirms completion. 6. The system records completed distraction task.
Alternative Flows	A1: Change Distraction Choice 3a. Quitter switches from walking to sound distraction. 3b. System activates sound player interface. 3c. Activity proceeds.
Quality Attributes	1. Interactivity: Immediate visual/auditory feedback upon user tap. 2. Usability: Simple one-tap logging.

6.10 UC-10 : Access Peer Support Group

Field	Description
Use Case No	UC-10
Use Case Name	Participate in Peer Support Group Chat
Goal	To allow tobacco quitters to share encouragement, progress, and mutual support in a public community chatroom.
Primary Actors	Tobacco Quitter, Peer Group Member
Secondary Actors	None
Precondition	N/A
Success End Condition	Group message is posted and visible in real-time to community members.
Failed End Condition	Message delivery fails due to empty message text.
Trigger	The quitter enters the Peer Support screen.
Main Success Flow	1. The quitter opens Peer Support Group Chat. 2. The system loads recent group messages. 3. The quitter types a supportive message. 4. The system validates text content. 5. The system posts message to

	backend chat endpoint. 6. The message appears in real-time chat feed.
Alternative Flows	A1: Empty Message 4a. System detects empty text string. 4b. Send button remains disabled. 4c. User enters text.
Quality Attributes	1. Performance: Messages shall sync within 1 second. 2. Security: Inappropriate content filtering.

6.11 UC-11 : View QuitMate - Tobacco Awareness

Field	Description
Use Case No	UC-11
Use Case Name	View Tobacco Awareness & Health Timeline
Goal	To provide educational content, health recovery milestones, and tobacco harm awareness articles.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Awareness articles and health recovery timeline cards are displayed.
Failed End Condition	Educational content fails to load.
Trigger	The quitter selects Education & Awareness section.
Main Success Flow	1. The quitter opens Tobacco Awareness section. 2. The system fetches awareness categories and recovery timeline (24h, 7d, 1yr health improvement). 3. The quitter selects a health article. 4. The system renders detailed article text and graphics.
Alternative Flows	A1: Offline Mode 2a. Device is offline. 2b. System loads offline bundled articles. 2c. Content displays seamlessly.
Quality Attributes	1. Usability: Typography shall be clear and readable. 2. Educational Value: Information based on health guidelines.

6.12 UC-12 : View Gamification Progress

Field	Description
Use Case No	UC-12
Use Case Name	View Gamification Progress & Unlock Badges
Goal	To track user badges and streak milestones, clean streak levels, and display unlocked achievement badges.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Streak Badges progress, current user level, and earned achievement badges are displayed.
Failed End Condition	Gamification stats fail to refresh.
Trigger	The quitter opens Gamification & Achievements screen.
Main Success Flow	1. The quitter opens Gamification screen. 2. The system calculates current user level, total Streak Badges, and unlocked badges. 3. The system renders level progress gauge and badge gallery. 4. The quitter reviews earned achievements and upcoming milestones.
Alternative Flows	A1: Badge Unlocked Event 2a. System detects criteria met for new badge (e.g. 7 Days Clean). 2b. System triggers celebratory dialog animation. 2c. Badge added to gallery.
Quality Attributes	1. Gamification Appeal: Visual badges and Streak Badges counters shall motivate user. 2. Accuracy: Streak Badges calculations match check-in logs.

6.13 UC-13 : Send Peer Support Messages

Field	Description
Use Case No	UC-13
Use Case Name	Interact with AI Craving Companion
Goal	To provide intelligent, personalized craving advice and motivational conversation using Groq AI Service integration.
Primary Actors	Tobacco Quitter

Secondary Actors	Groq AI Service
Precondition	N/A
Success End Condition	AI companion responds with helpful craving coping strategies.
Failed End Condition	AI response fails due to API timeout.
Trigger	The quitter sends a query to the AI Assistant.
Main Success Flow	1. The quitter opens AI Craving Companion tab. 2. The quitter types a craving query or feeling. 3. The system sends prompt payload to Groq AI Service API endpoint. 4. Groq AI generates intelligent response. 5. System renders response in chat view.
Alternative Flows	A1: AI Service Timeout 4a. Groq AI API times out. 4b. System displays fallback motivational response. 4c. Quitter receives immediate guidance.
Quality Attributes	1. Intelligence: Responses tailored to tobacco cessation context. 2. Performance: AI reply rendered within 2.5 seconds.

6.14 UC-14 : Receive Peer Support Messages

Field	Description
Use Case No	UC-14
Use Case Name	Receive Peer Support Group Messages
Goal	To receive real-time community chat messages posted by other group members.
Primary Actors	Tobacco Quitter, Peer Group Member
Secondary Actors	None
Precondition	N/A
Success End Condition	Incoming peer group messages are delivered and appended to chat feed.
Failed End Condition	Incoming chat payload fails to parse.

Trigger	Another community member posts a message to group chat.
Main Success Flow	1. Community member posts message. 2. Backend chat server receives payload. 3. System broadcasts message to connected app clients. 4. App receives payload and updates UI stream. 5. Quitter reads new message.
Alternative Flows	A1: App in Background 4a. App is minimized. 4b. System issues subtle local notification. 4c. User opens app to view message.
Quality Attributes	1. Real-time: Sub-second message delivery. 2. Reliability: No dropped message events.

6.15 UC-15 : Offline Data Caching

Field	Description
Use Case No	UC-15
Use Case Name	Offline Data Caching & Auto-Synchronization
Goal	To cache check-ins, profile edits, and savings goals locally when offline, and auto-sync with database when connection returns.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Local changes stored in Hive offline cache are synced to PostgreSQL database upon reconnection.
Failed End Condition	Sync fails due to persistent network failure.
Trigger	App detects restored internet connection or user performs action while offline.
Main Success Flow	1. Quitter performs app action (check-in, goal creation) offline. 2. System saves data payload to local Hive storage box. 3. System monitors network status. 4. Internet connection is restored. 5. System initiates background sync service. 6. System sends cached payloads to PostgreSQL backend. 7. Server confirms successful database update.
Alternative Flows	A1: Conflict Resolution 6a. Server detects version conflict.

	6b. Server resolves using latest timestamp. 6c. Database updated cleanly.
Quality Attributes	1. Seamlessness: Sync runs silently in background. 2. Integrity: Zero data loss during offline operation.

6.16 UC-16 : Send Notification

Field	Description
Use Case No	UC-16
Use Case Name	Receive Local & Push Notifications
Goal	To deliver daily check-in reminders, motivational quotes, and survival alerts to keep quitter engaged.
Primary Actors	Tobacco Quitter
Secondary Actors	None
Precondition	N/A
Success End Condition	Notification alert is displayed on device and opens relevant app screen when tapped.
Failed End Condition	Notification permission disabled by user.
Trigger	Scheduled daily check-in time or system trigger event occurs.
Main Success Flow	1. System schedules check-in reminder event. 2. Local notification manager triggers alert at scheduled time. 3. Device presents banner alert to quitter. 4. Quitter taps notification. 5. App opens directly to Daily Check-in screen.
Alternative Flows	A1: Permission Denied 2a. Notification permissions disabled. 2b. App falls back to in-app banner badges. 2c. Quitter sees badge on next launch.
Quality Attributes	1. Timeliness: Notifications delivered precisely on schedule. 2. Usability: Tapping banner routes to exact target screen.

6.17 UC-17 : Manage Content

Field	Description
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Use Case No	UC-17
Use Case Name	Educational Content Management
Goal	To allow system administrator to publish, edit, or remove tobacco awareness articles and health resources.
Primary Actors	System Administrator
Secondary Actors	None
Precondition	N/A
Success End Condition	Content article is created or updated in database and available to all app users.
Failed End Condition	Publishing fails due to missing mandatory title or body text.
Trigger	System administrator accesses Web Admin Portal content section.
Main Success Flow	1. Administrator logs into Web Admin Portal. 2. Administrator opens Content Management tab. 3. Administrator fills article title, category, and markdown body. 4. System validates required fields. 5. System saves article to PostgreSQL database. 6. Article becomes visible in mobile app.
Alternative Flows	A1: Validation Failure 4a. Missing title or body. 4b. System highlights required input box. 4c. Admin fills missing content.
Quality Attributes	1. Security: Requires admin authorization token. 2. Usability: Rich text editor interface.

6.18 UC-18 : System Administration & Community Moderation

Field	Description
Use Case No	UC-18
Use Case Name	System Administration & Community Moderation
Goal	To allow system administrator to review community chat messages, moderate inappropriate content, and manage user status.
Primary Actors	System Administrator

Secondary Actors	None
Precondition	N/A
Success End Condition	Moderation action is executed and database audit log is updated.
Failed End Condition	Action cancelled by admin.
Trigger	Administrator reviews reported messages or user management tab.
Main Success Flow	1. Administrator accesses Moderation Control Panel. 2. Administrator views list of chat messages and registered user accounts. 3. Administrator selects an inappropriate message or user record. 4. Administrator executes delete message or flag account action. 5. System updates database tables and updates live chatroom.
Alternative Flows	A1: Undo Action 4a. Admin cancels moderation dialog. 4b. Action is aborted. 4c. No database change occurs.
Quality Attributes	1. Control: Instant moderation execution. 2. Auditability: All admin actions recorded in audit log.

7 Activity Diagrams

7.1 Activity Diagram for Manage Profile

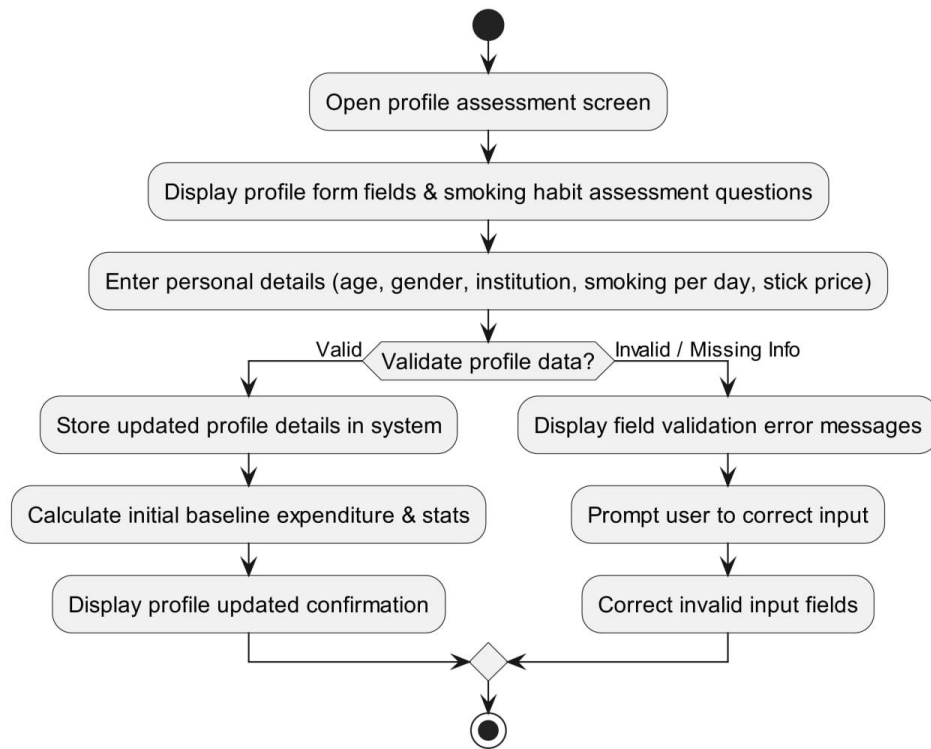


Figure 2: Activity Diagram for Manage Profile

7.2 Activity Diagram for Set Quit Date

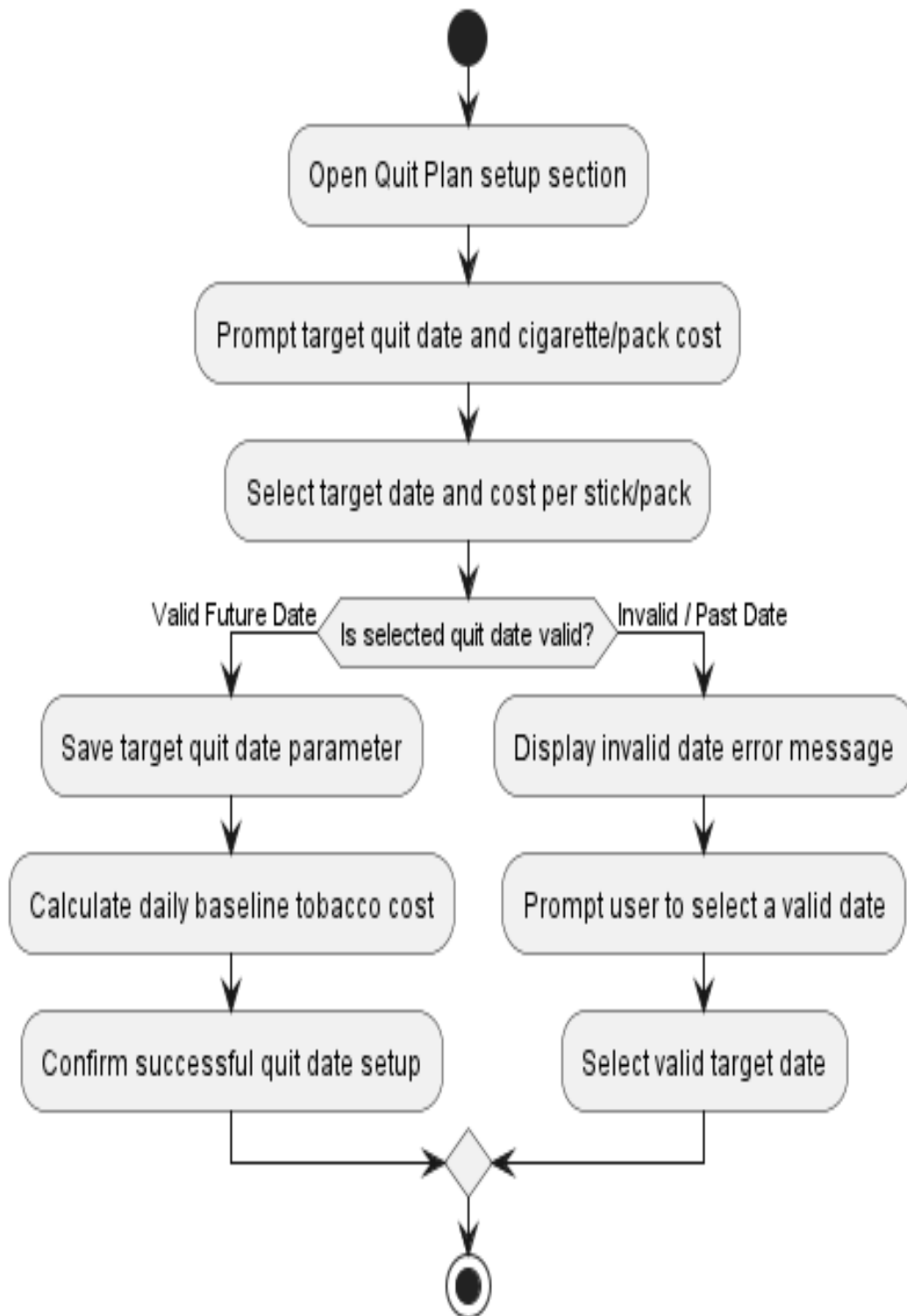


Figure 3: Activity Diagram for Set Quit Date

7.3 Activity Diagram for View Quit Plan

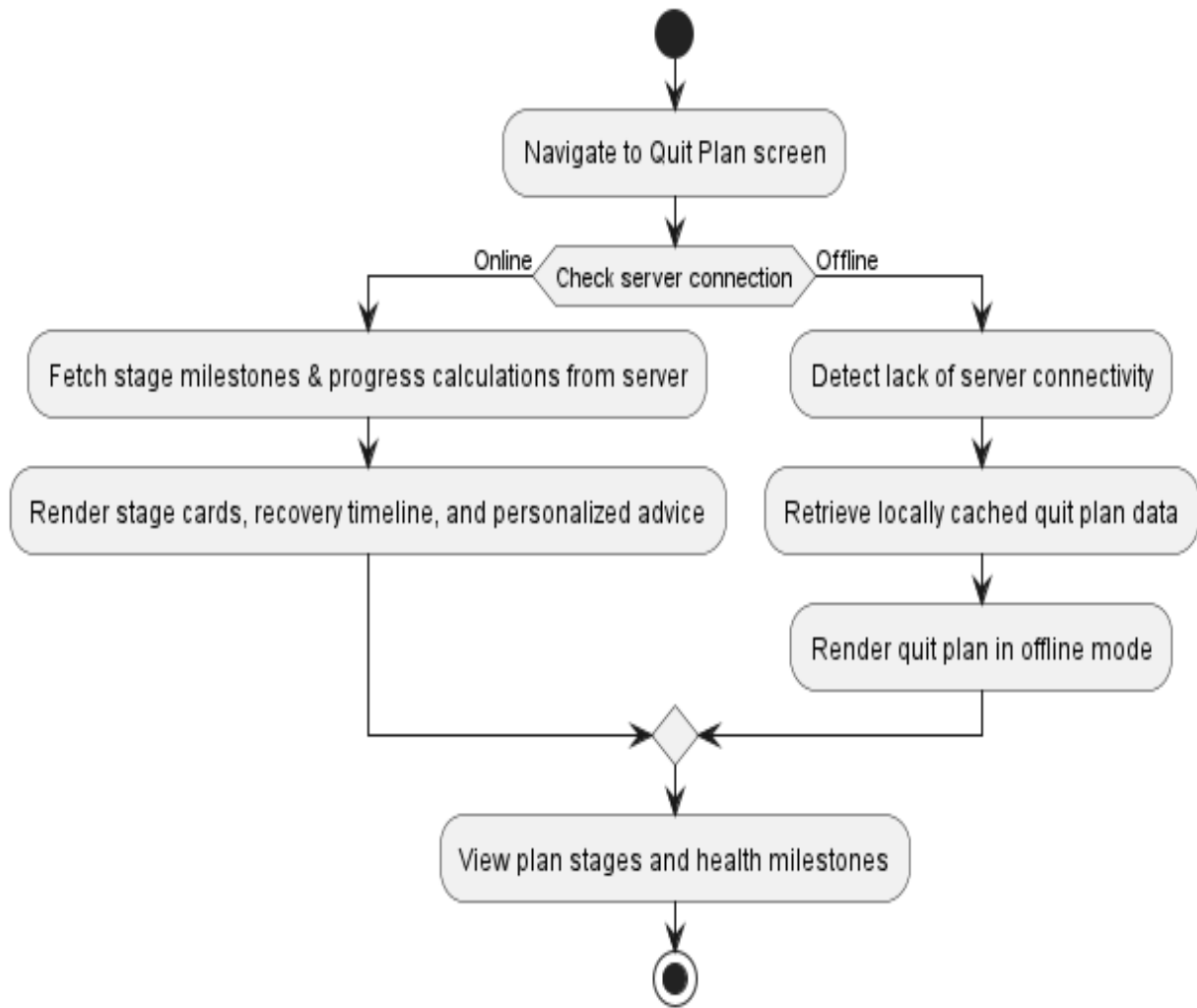


Figure 4: Activity Diagram for View Quit Plan

7.4 Activity Diagram for Daily Check-in

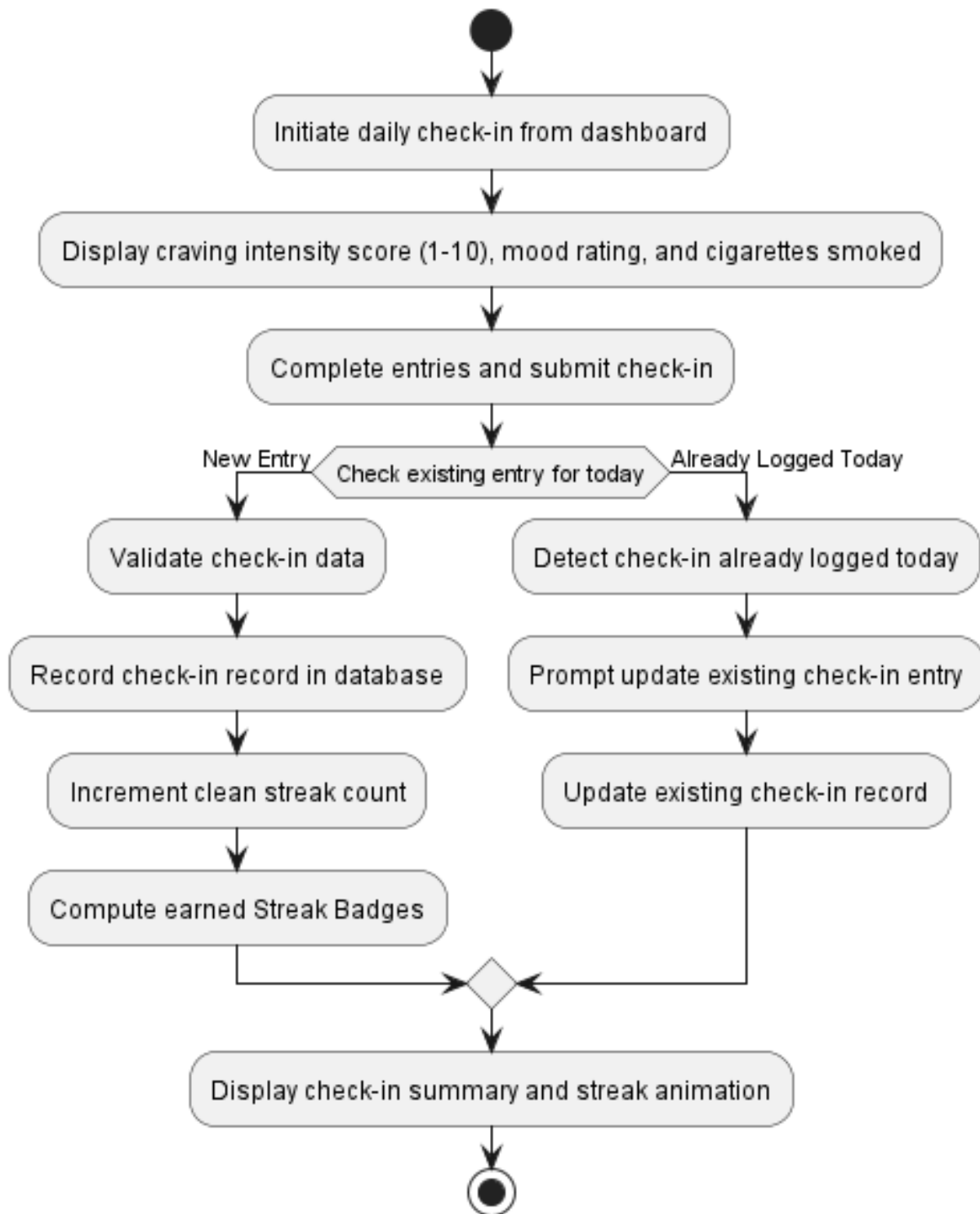


Figure 5: Activity Diagram for Daily Check-in

7.5 Activity Diagram for Track Saving

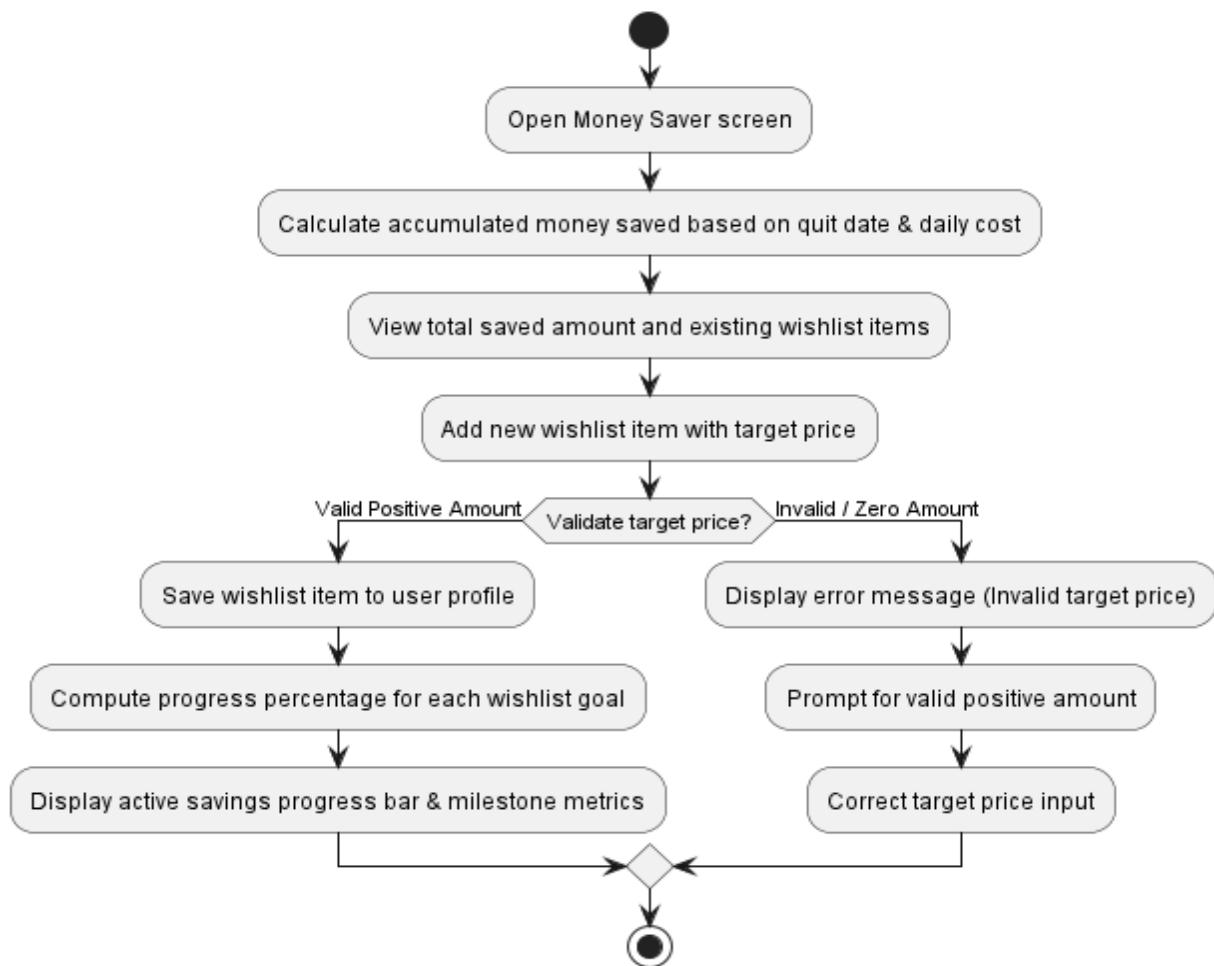


Figure 6: Activity Diagram for Track Saving

7.6 Activity Diagram for Craving Support

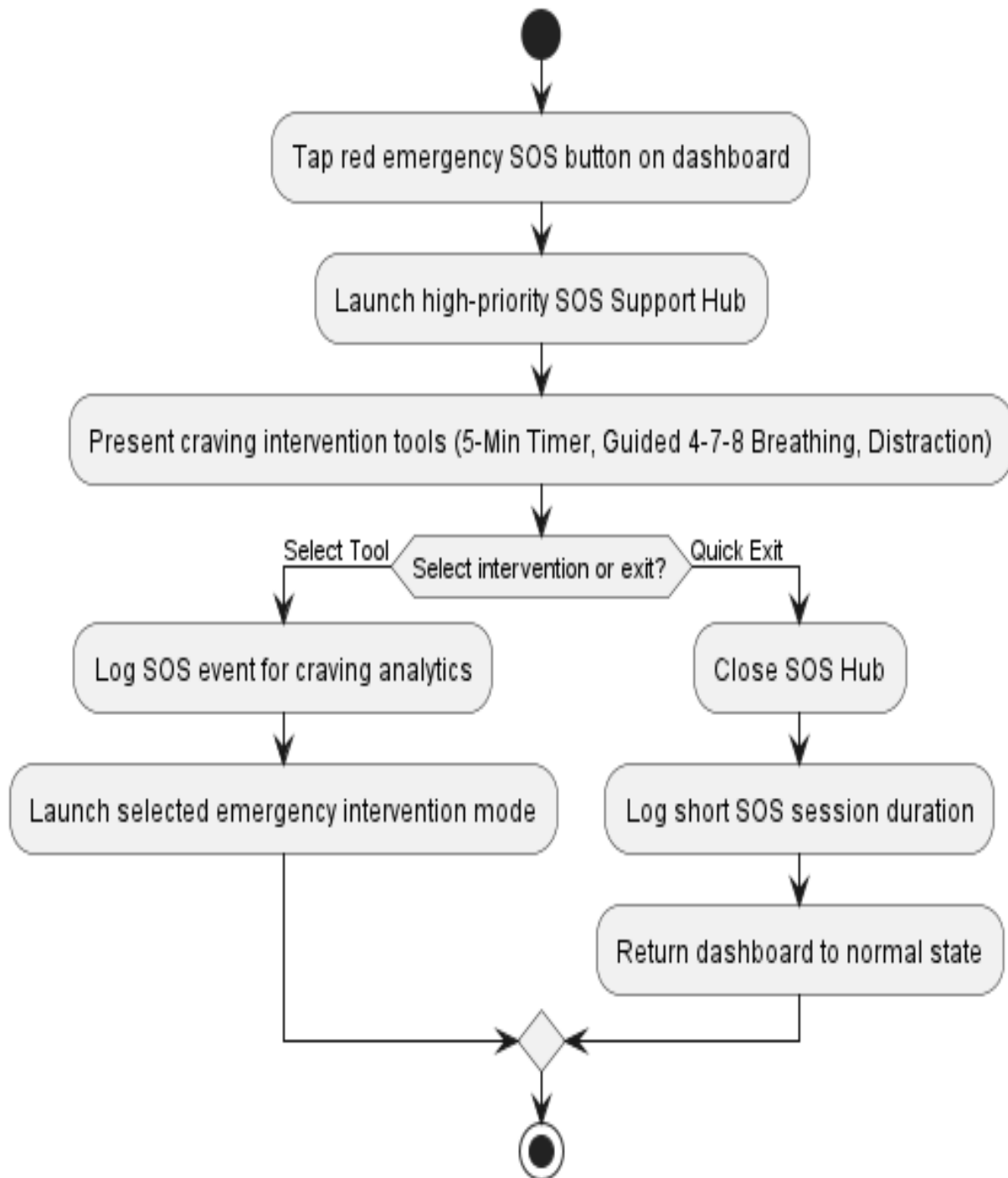


Figure 7: Activity Diagram for Craving Support

7.7 Activity Diagram for Start Survival Timer

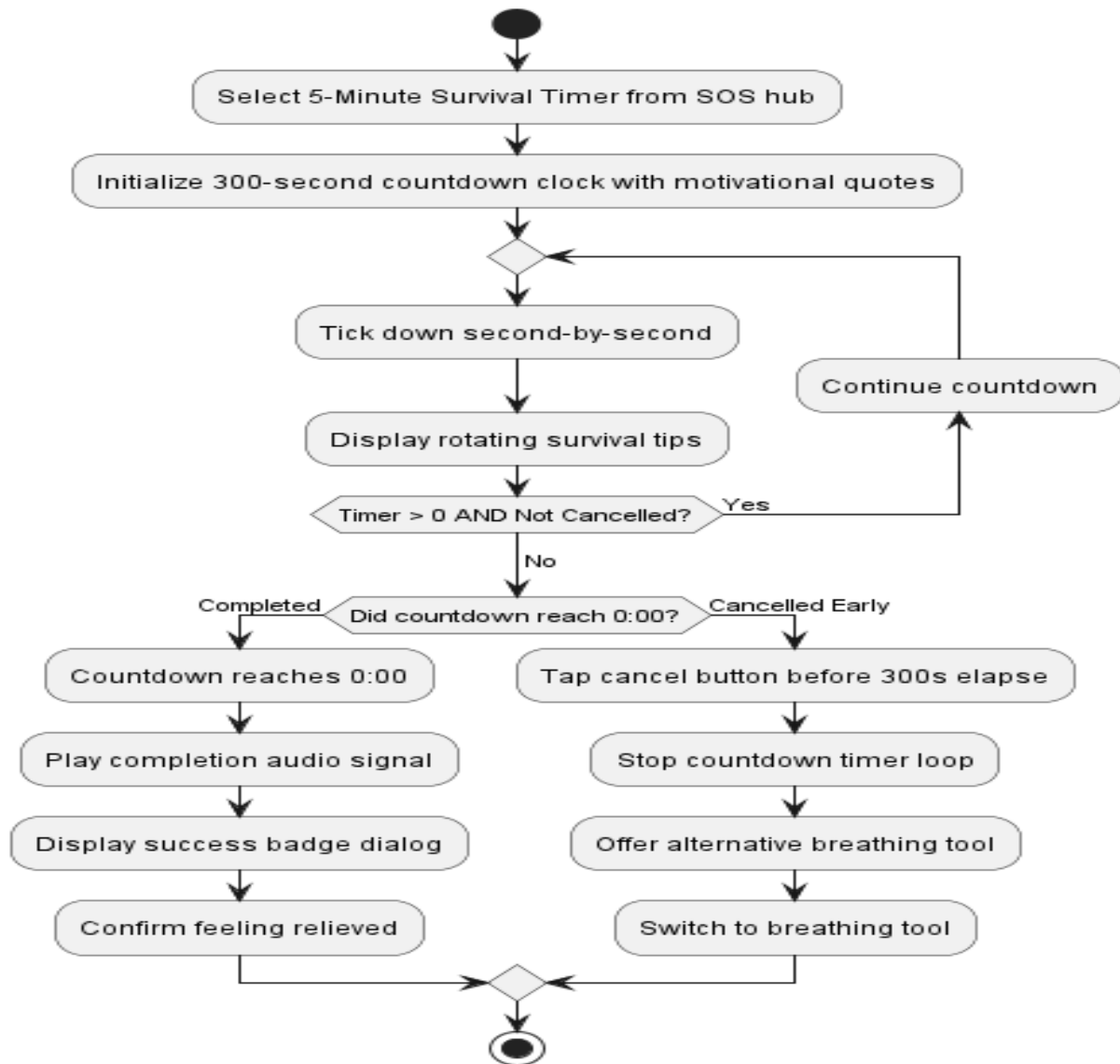


Figure 8: Activity Diagram for Start Survival Timer

7.8 Activity Diagram for Breathing Exercise

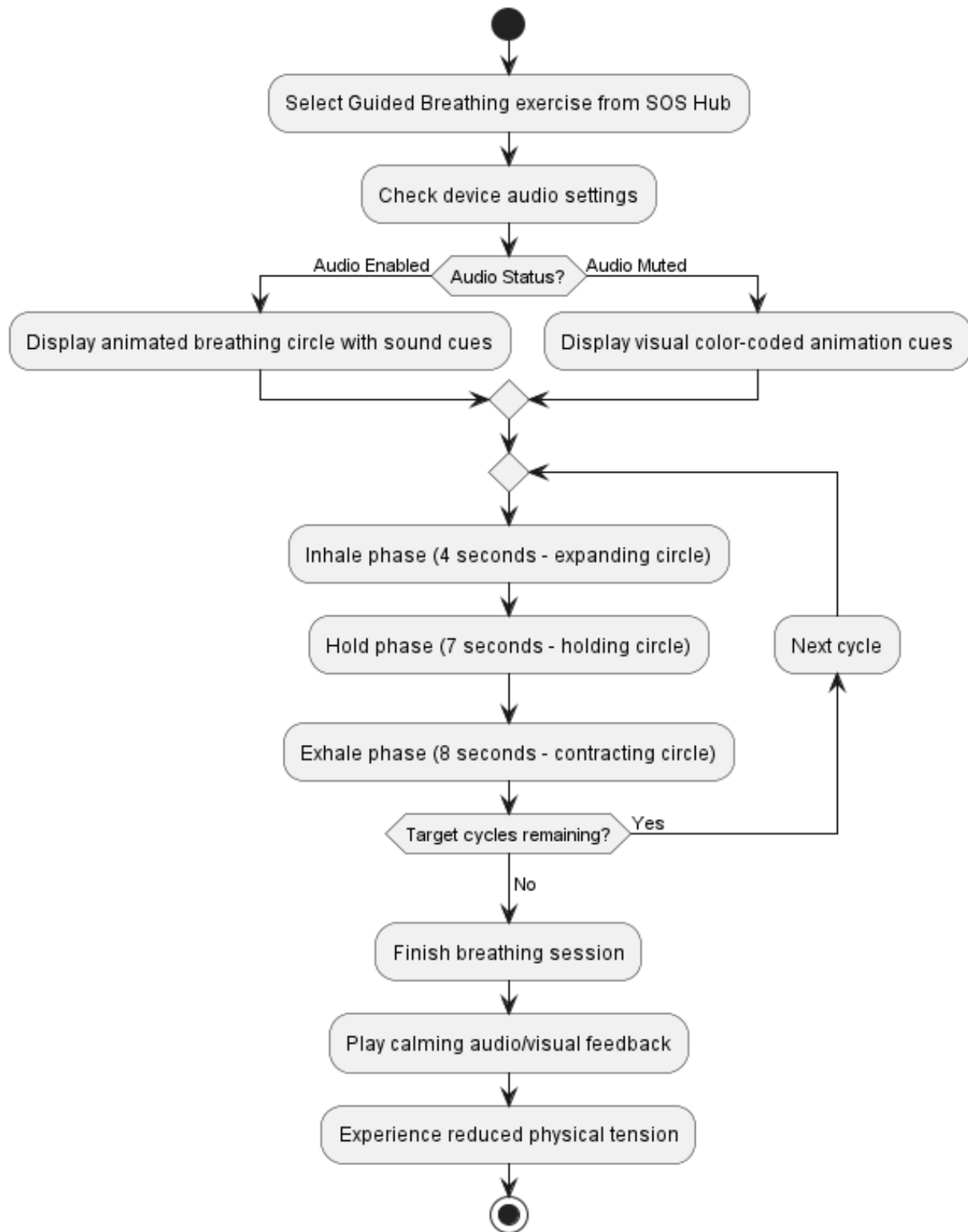


Figure 9: Activity Diagram for Breathing Exercise

7.9 Activity Diagram for Quick Distraction

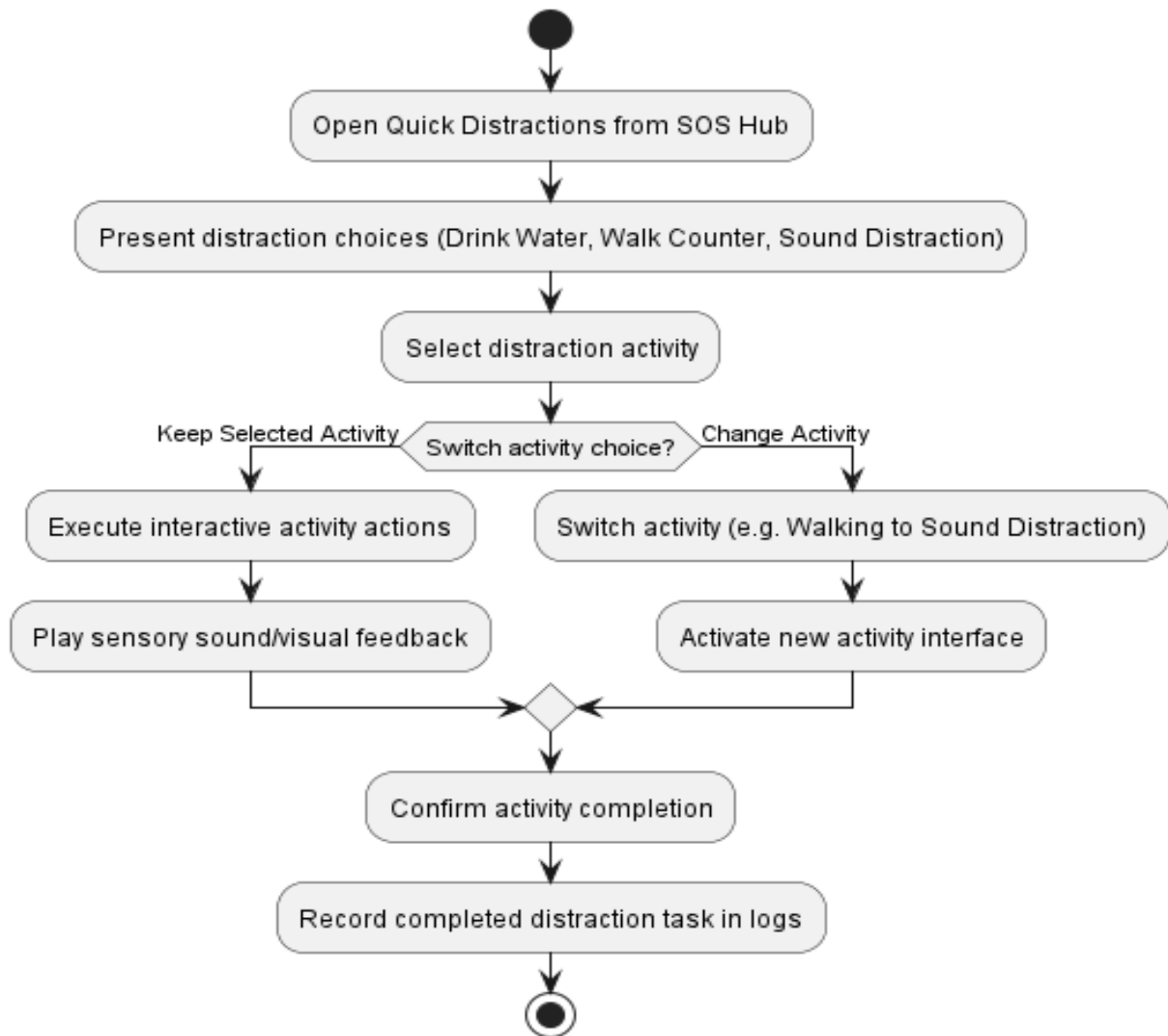


Figure 10: Activity Diagram for Quick Distraction

7.10 Activity Diagram for Access Peer Support Group

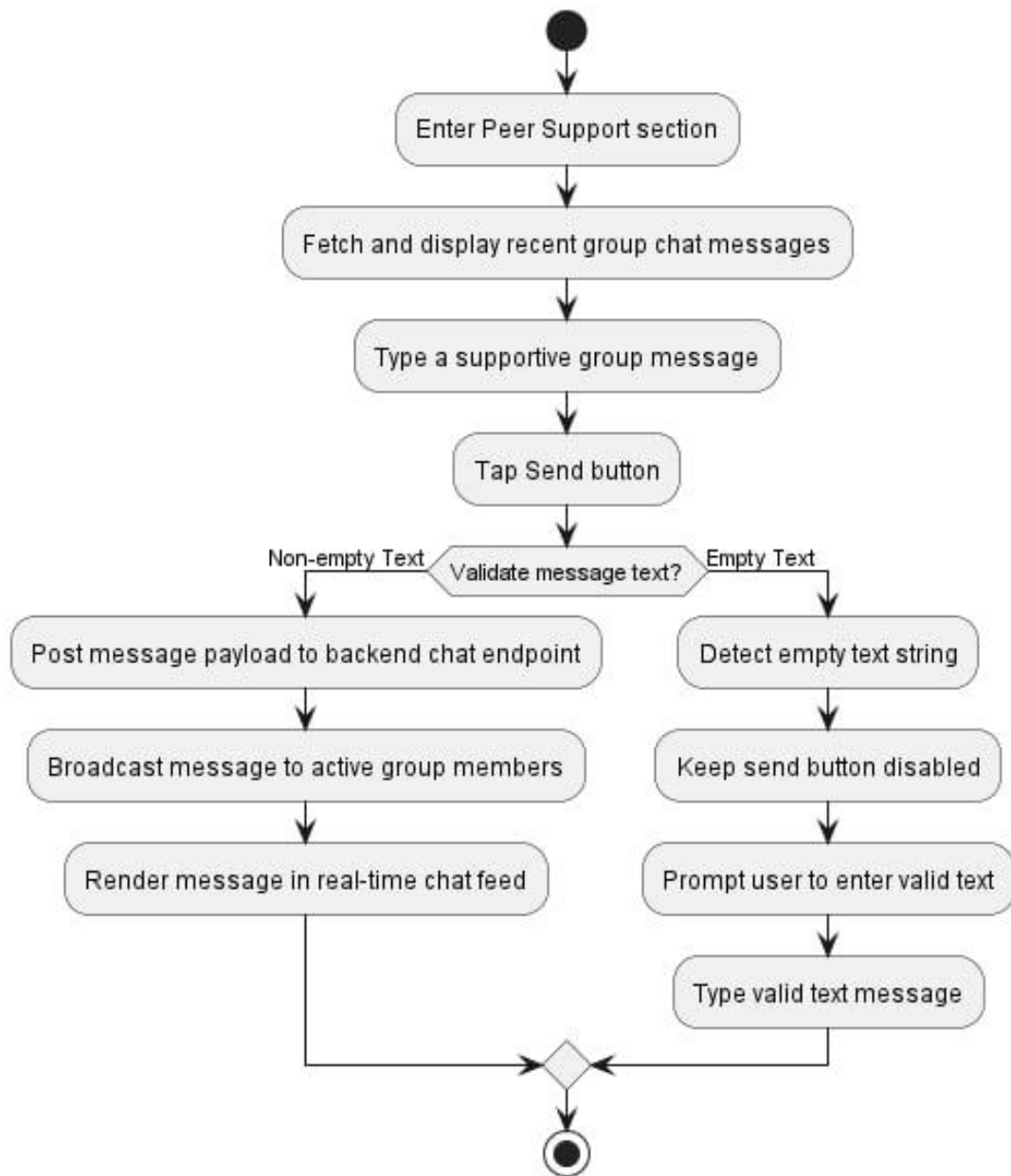


Figure 11: Activity Diagram for Access Peer Support Group

8. Sequence Diagram

8.1 Sequence Diagram for **Manage Profile**

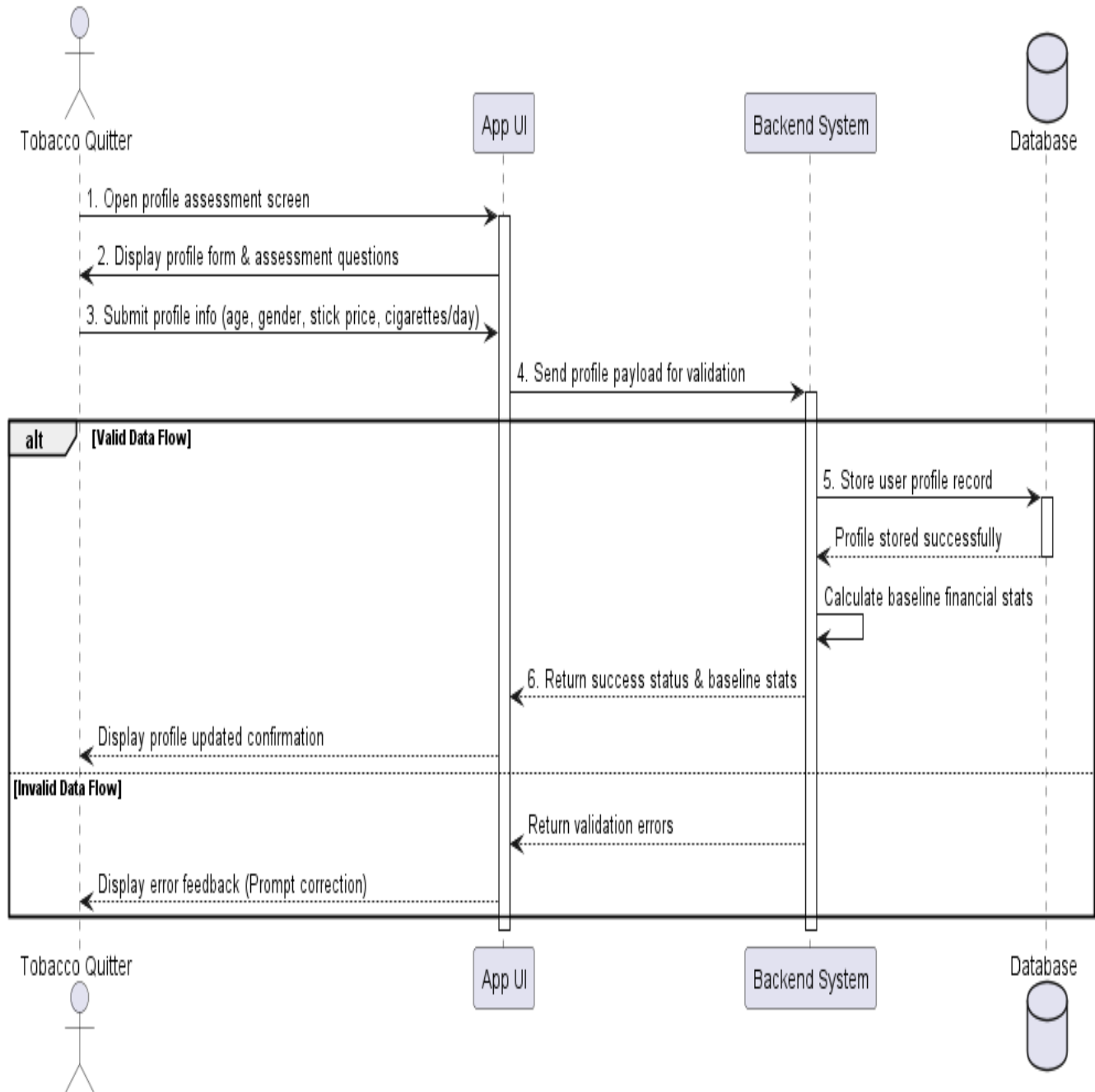


Figure 12: Sequence Diagram for Manage Profile

8.2 Sequence Diagram for Set Quit Date

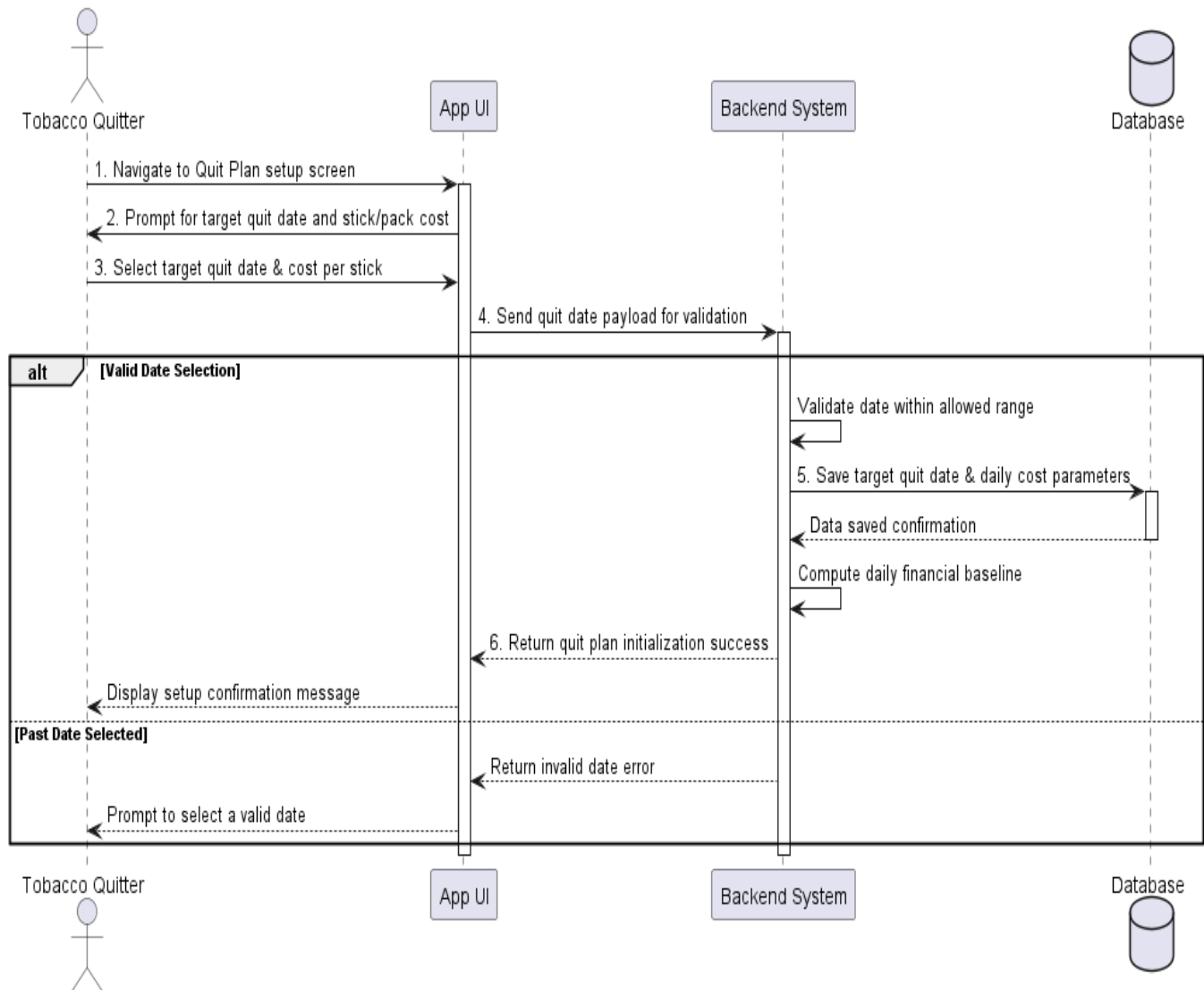


Figure 13: Sequence Diagram for Set Quit Date

8.3 Sequence Diagram for View Quit Plan

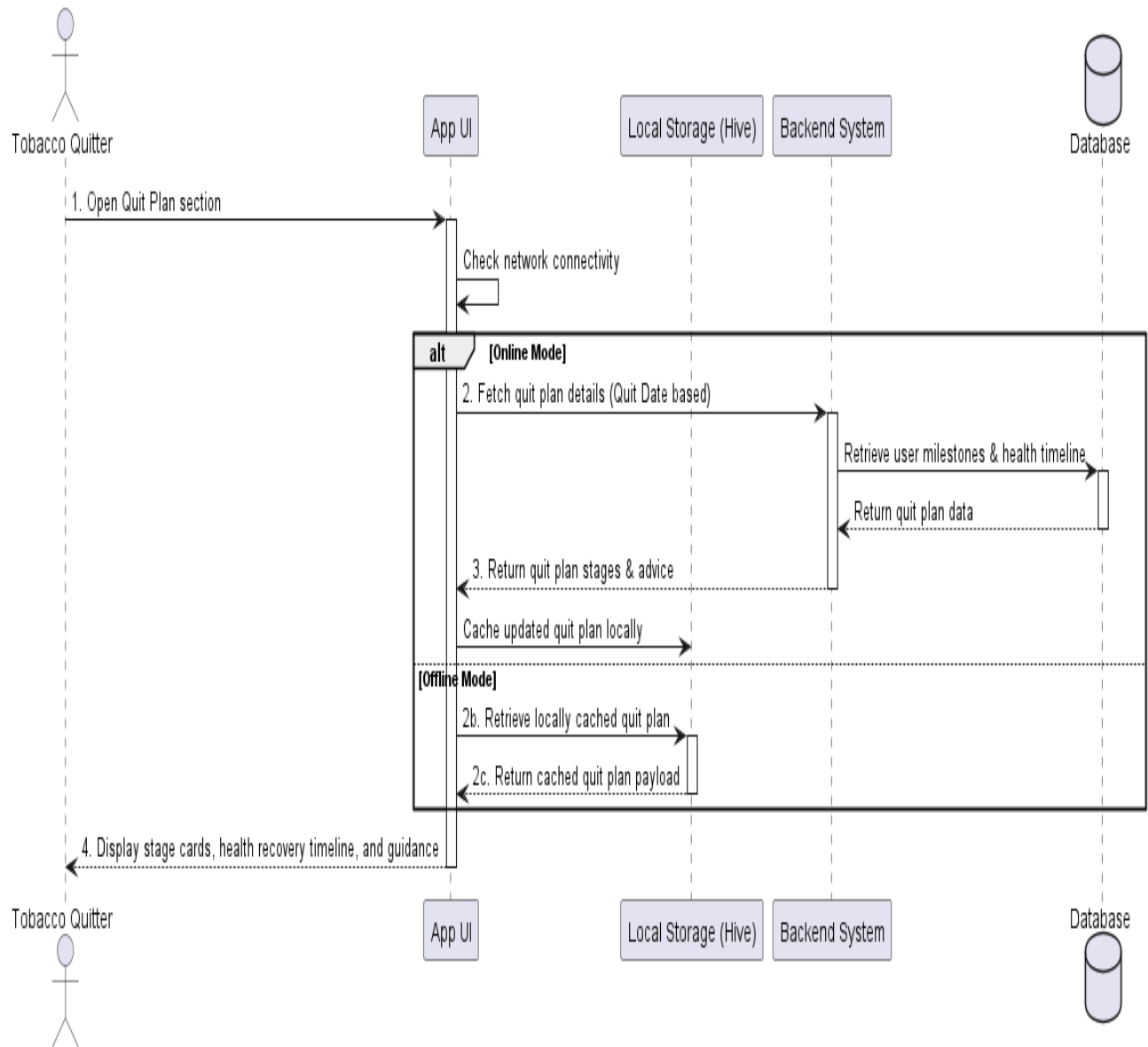


Figure 14: Sequence Diagram for View Quit Plan

8.4 Sequence Diagram for Daily Check-in

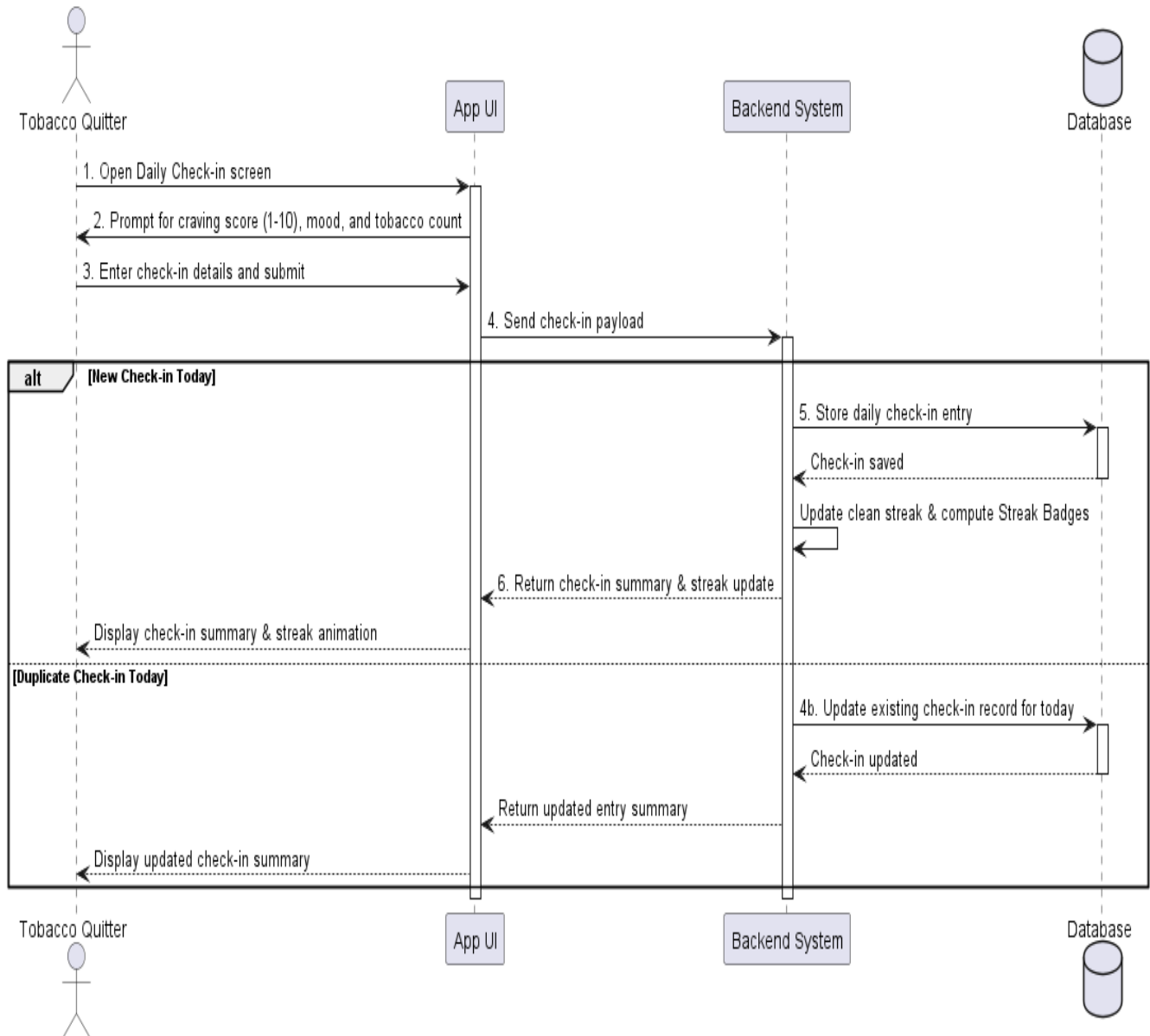


Figure 15: Sequence Diagram for Daily Check-in

8.5 Sequence Diagram for Track Saving

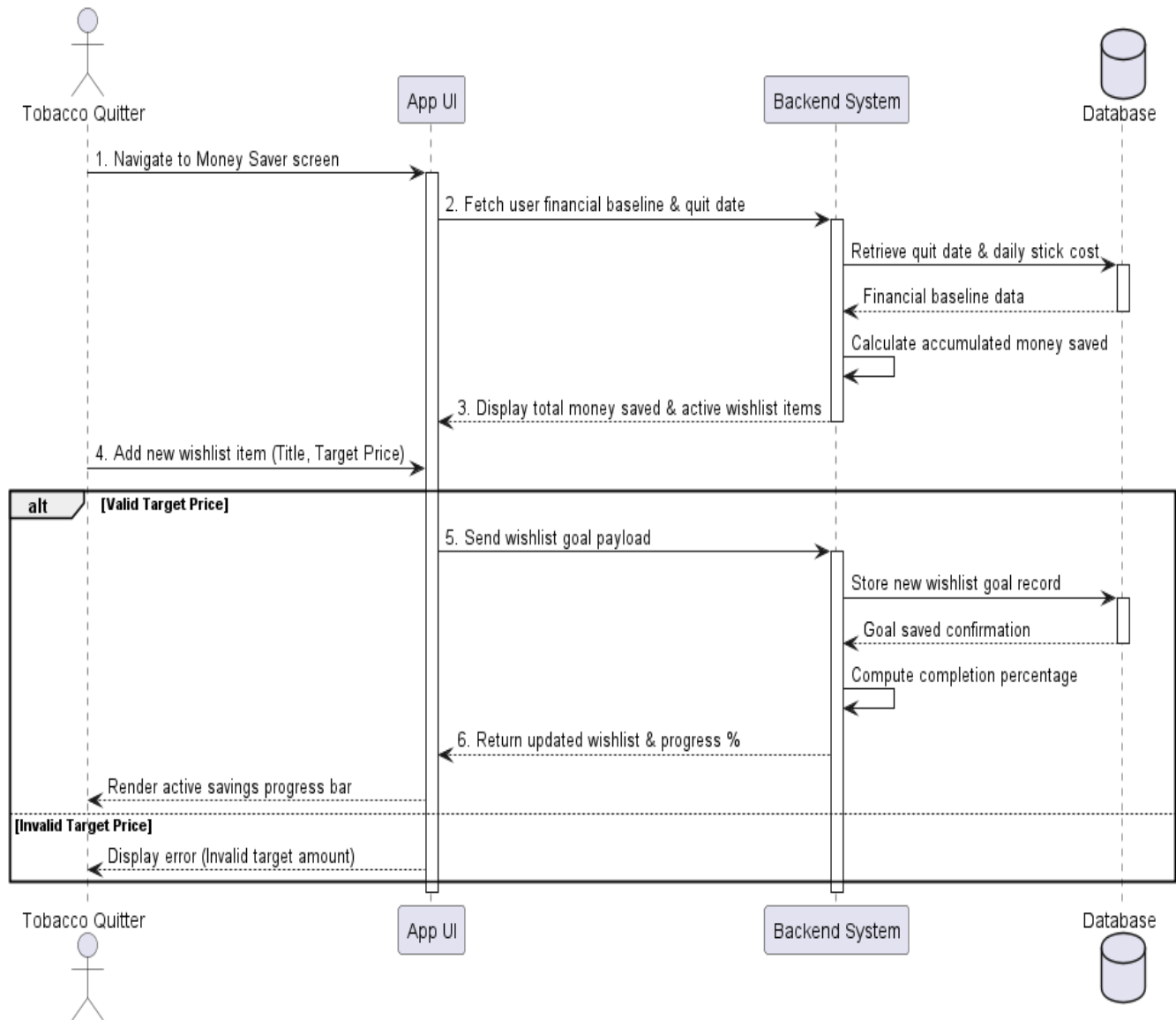


Figure 16: Sequence Diagram for Track Saving

8.6 Sequence Diagram for Craving Support

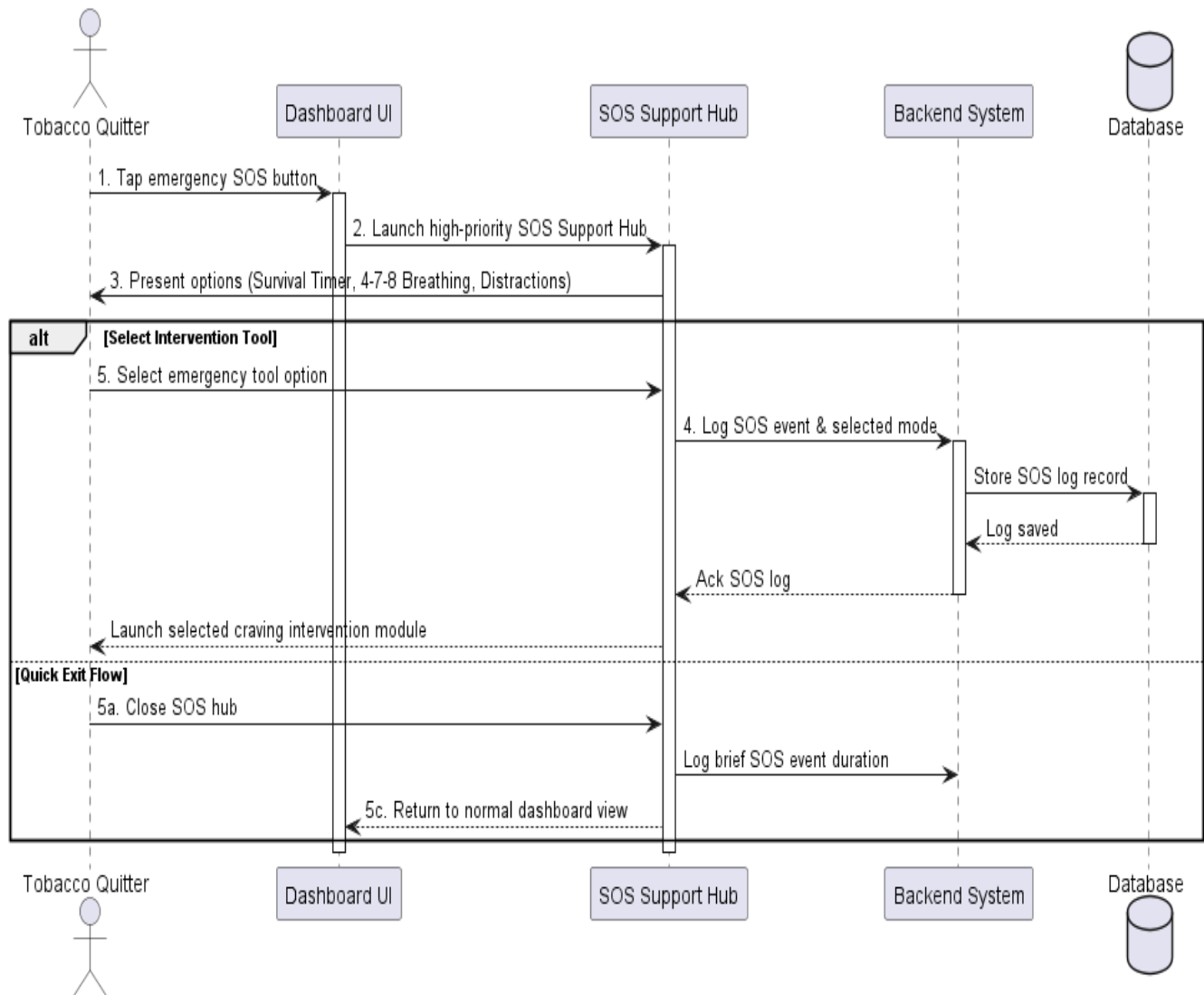


Figure 17: Sequence Diagram for Craving Support

8.7 Sequence Diagram for Start Survival Timer

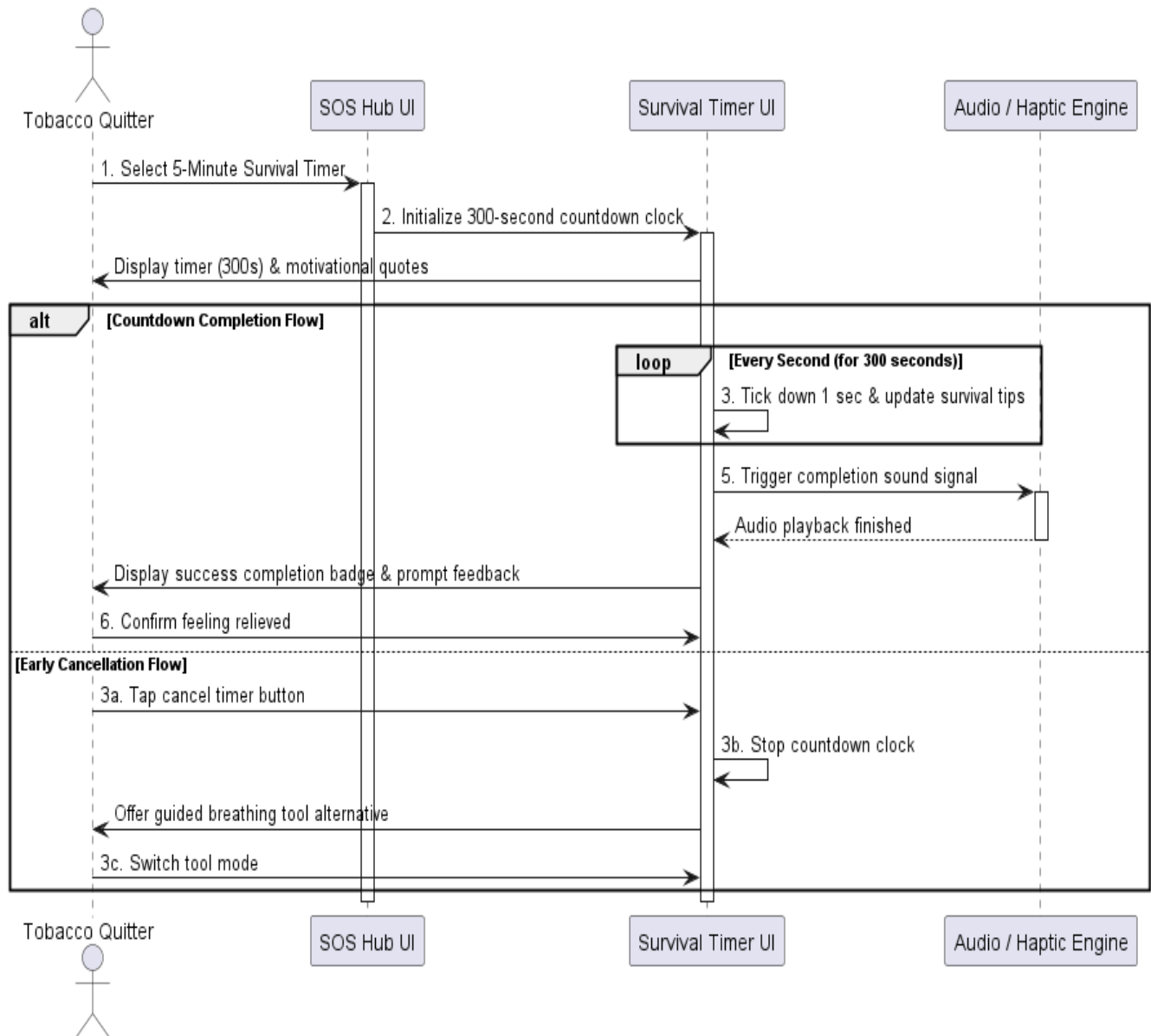


Figure 18: Sequence Diagram for Start Survival Timer

8.8 Sequence Diagram for Breathing Exercise

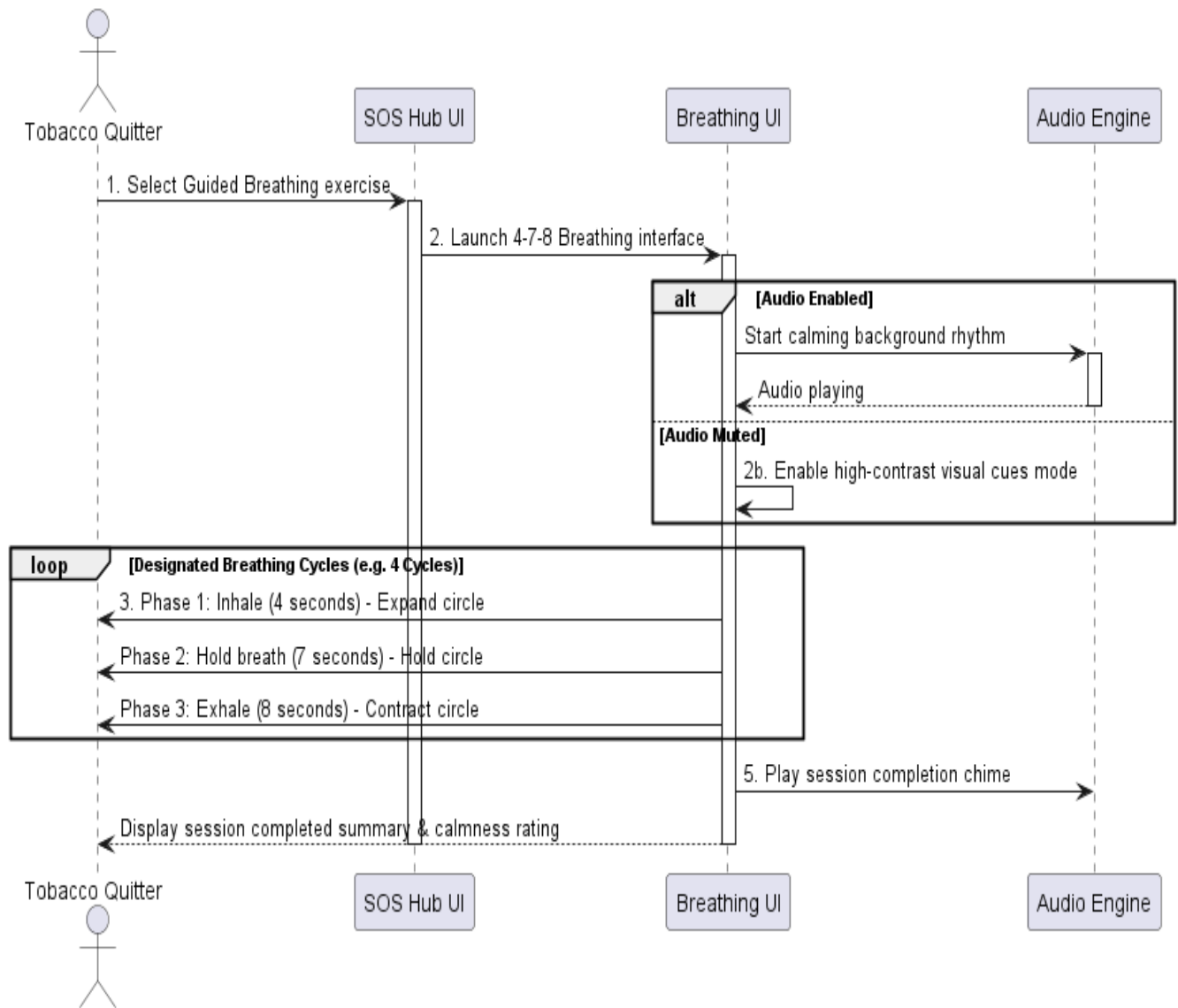


Figure 19: Sequence Diagram for Breathing Exercise

8.9 Sequence Diagram for Quick Distraction

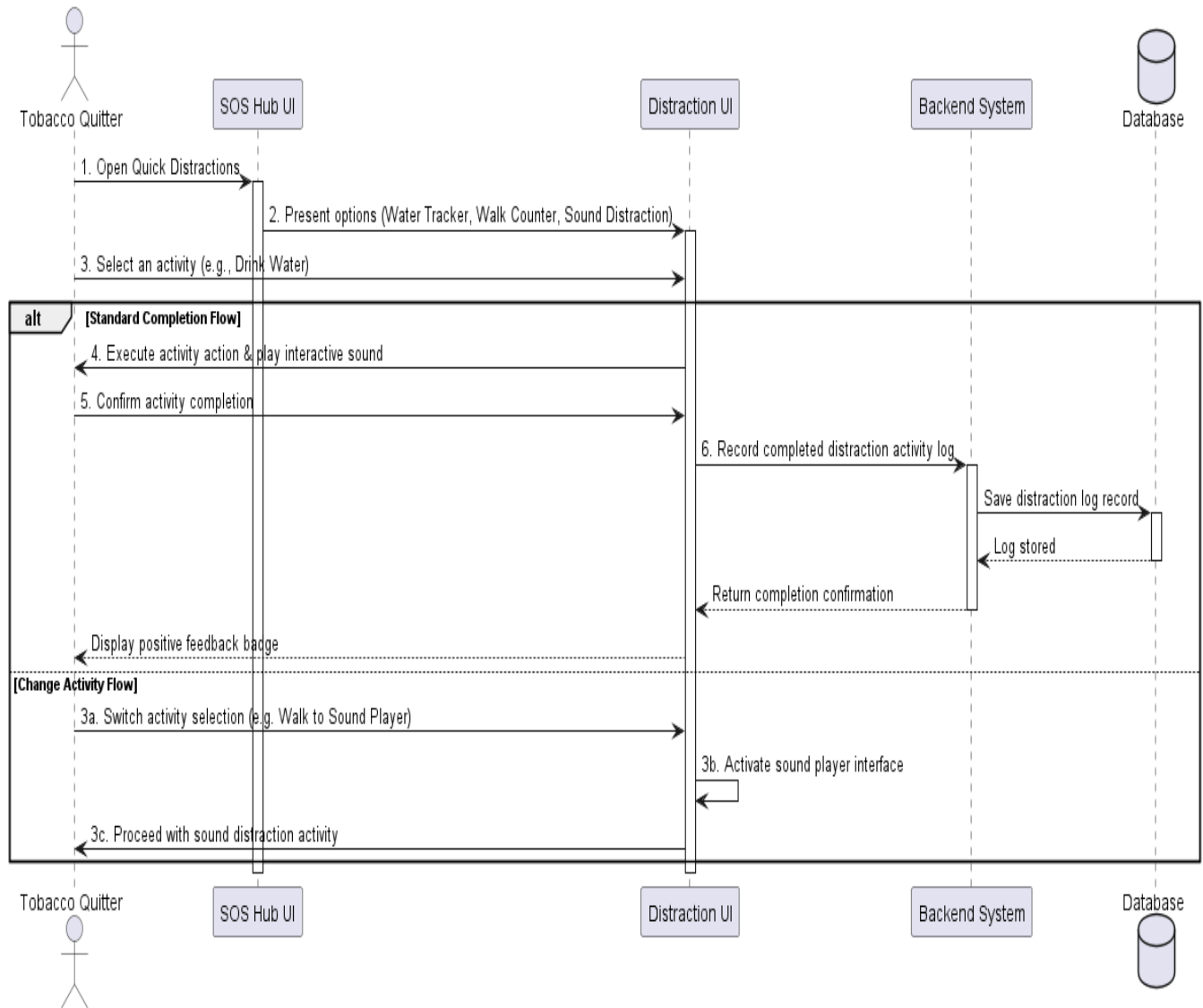


Figure 20: Sequence Diagram for Quick Distraction

8.10 Sequence Diagram for Access Peer Support Group

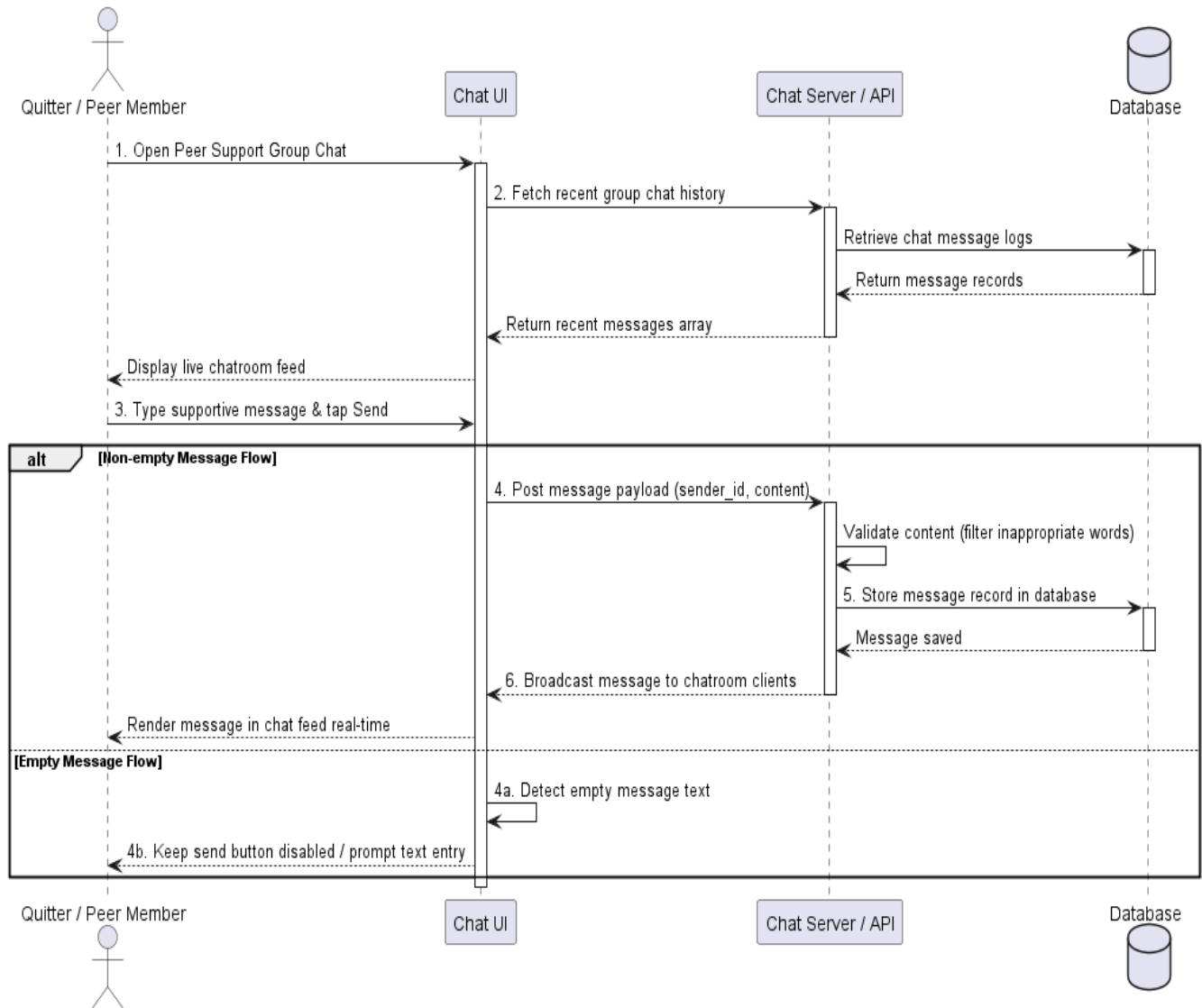


Figure 21: Sequence Diagram for Access Peer Support Group

9 Swimlane Diagram

9.1 Swimlane Diagram for Manage Profile

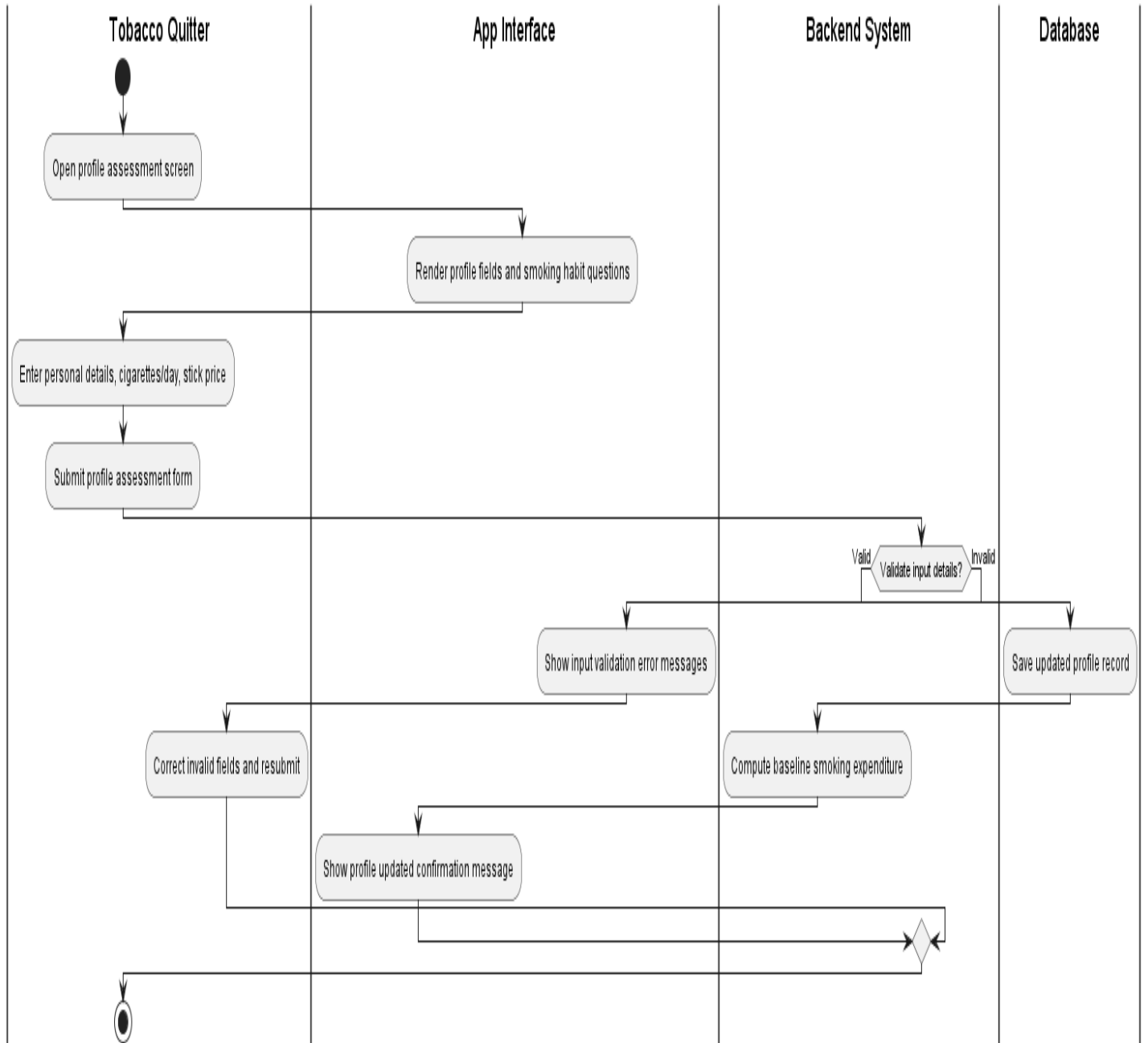


Figure 22: Swimlane Diagram for Manage Profile

9.2 Swimlane Diagram for Set Quit Date

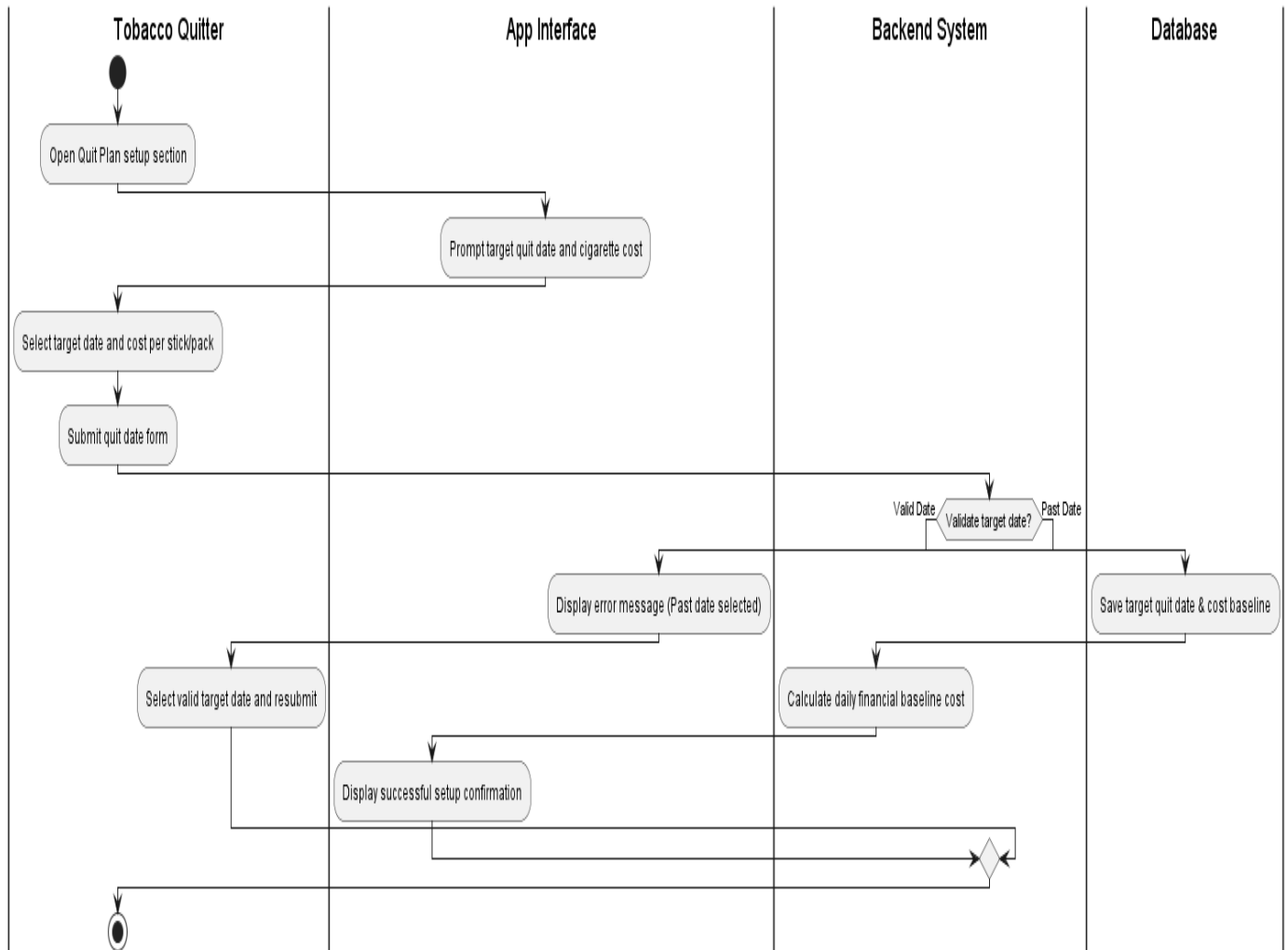


Figure 23: Swimlane Diagram for Set Quit Date

9.3 Swimlane Diagram for **View Quit Plan**

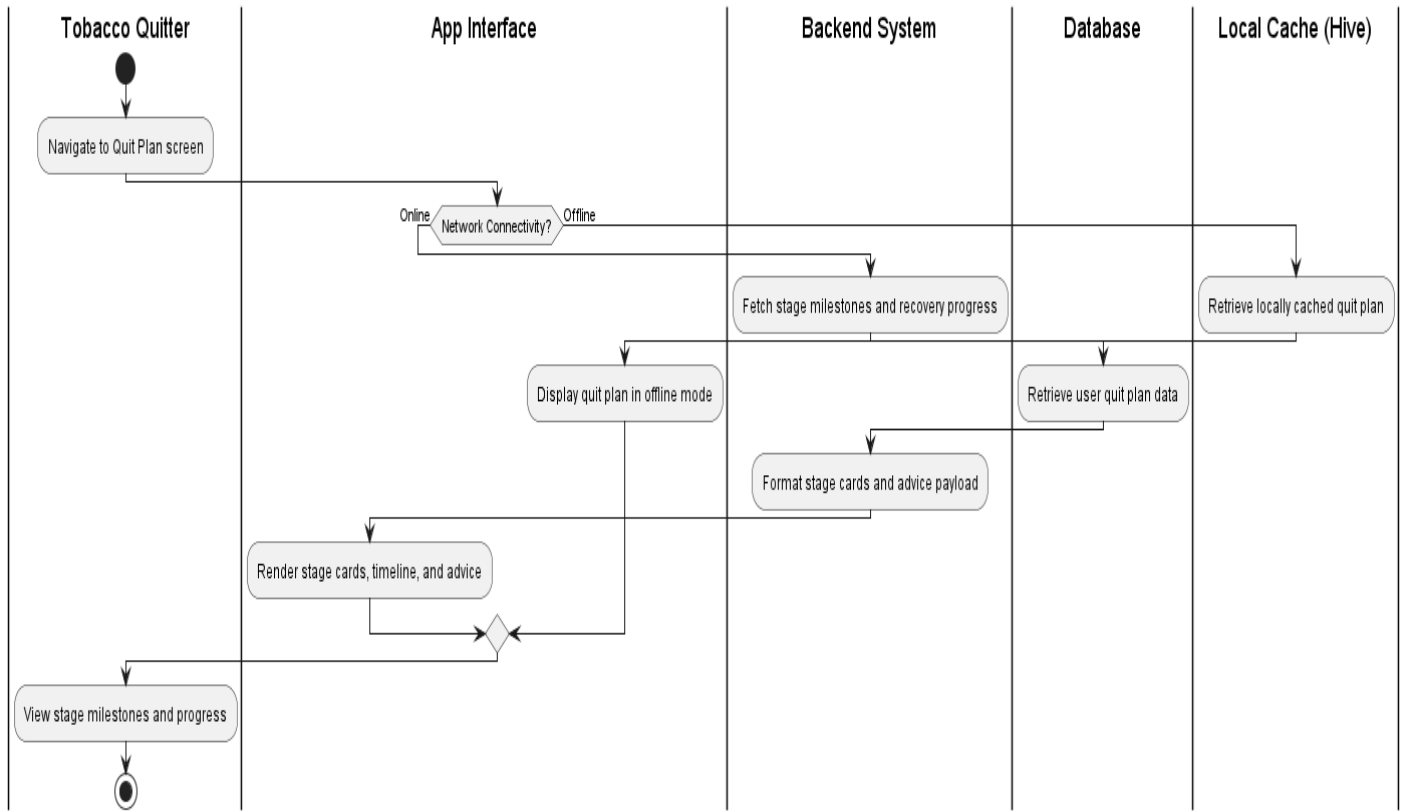


Figure 24: Swimlane Diagram for View Quit Plan

9.4 Swimlane Diagram for Daily Check-in

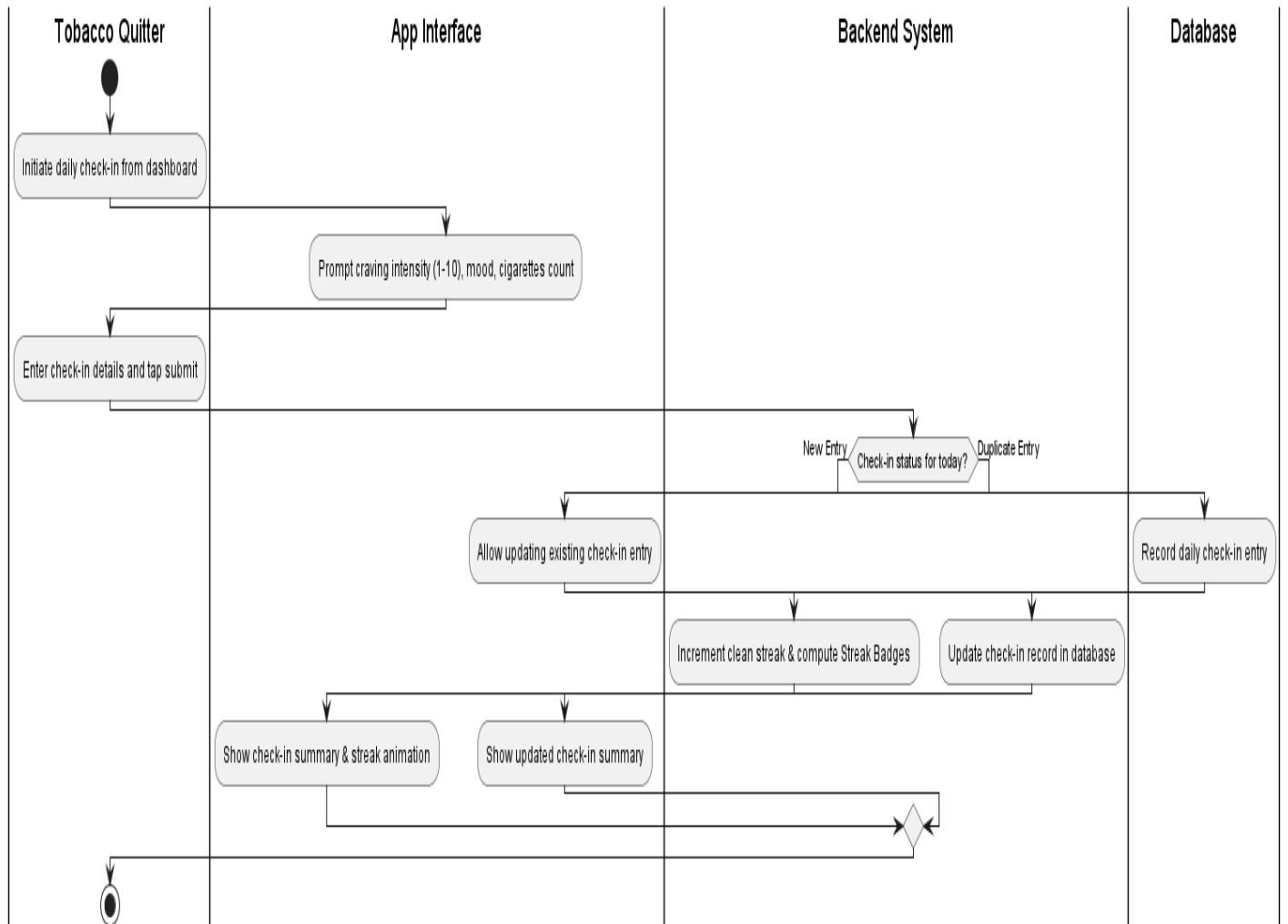


Figure 25: Swimlane Diagram for Daily Check-in

9.5 Swimlane Diagram for Track Saving

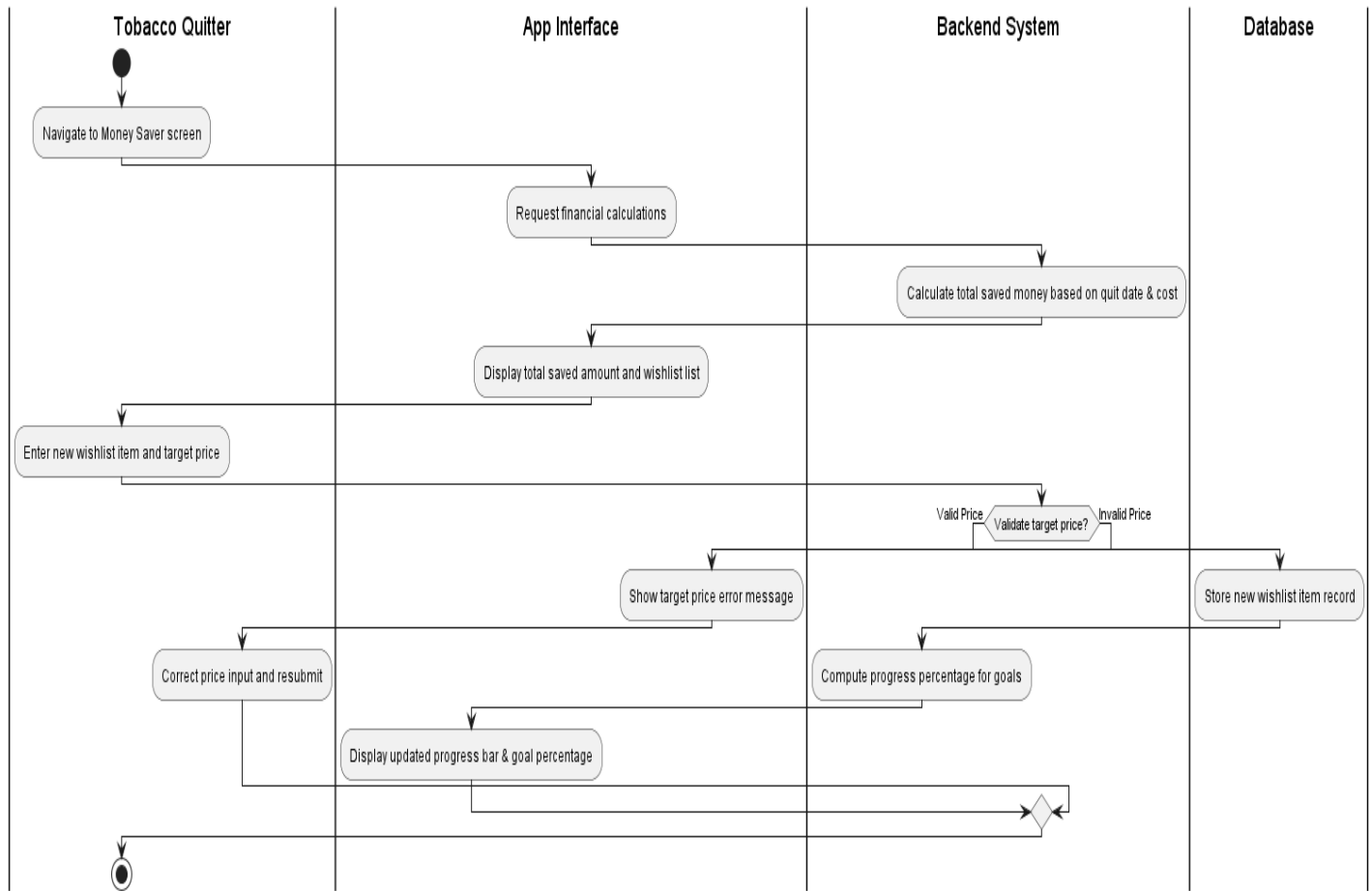


Figure 26: Swimlane Diagram for Track Saving

9.6 Swimlane Diagram for Craving Support

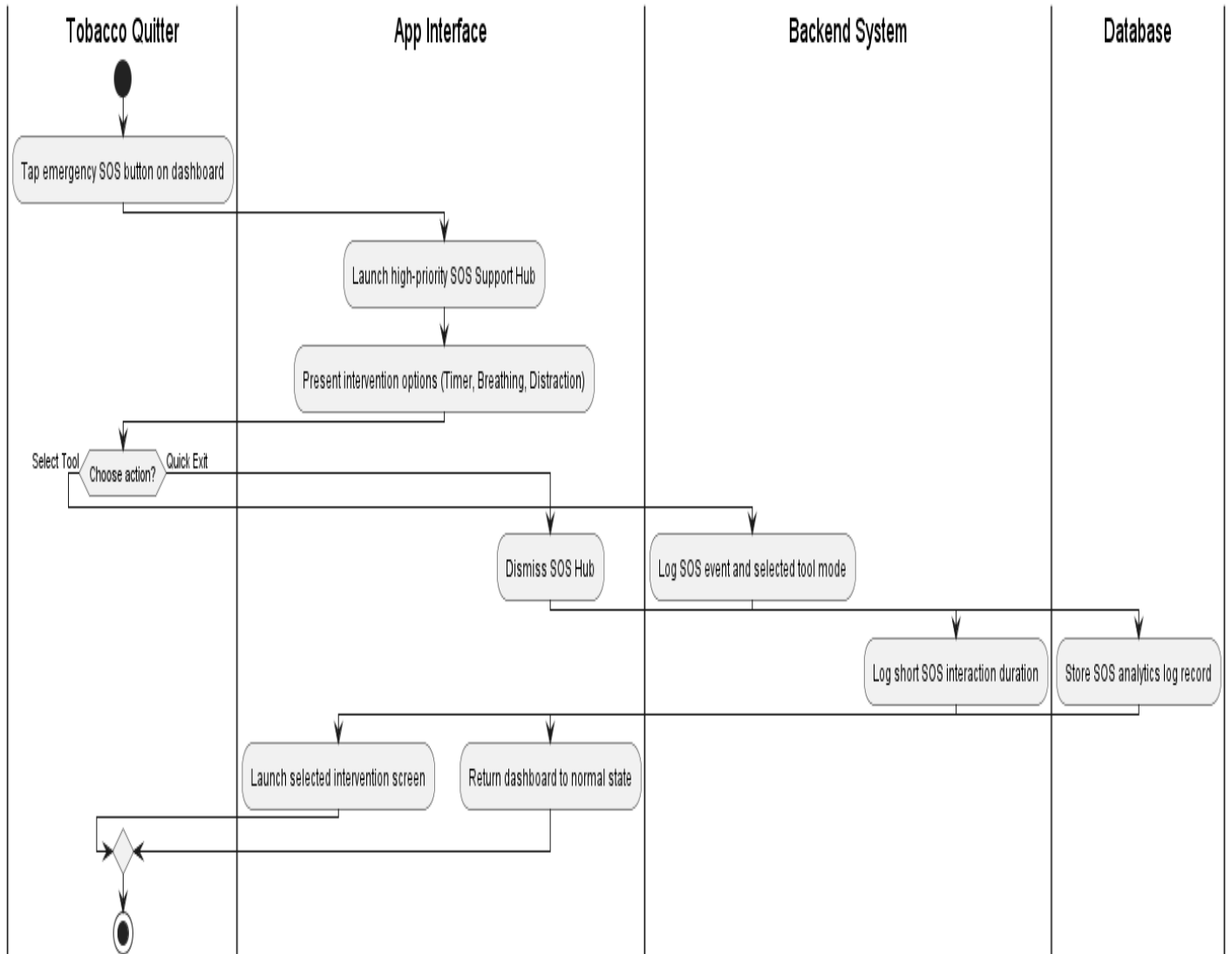


Figure 27: Swimlane Diagram for Craving Support

9.7 Swimlane Diagram for Start Survival Timer

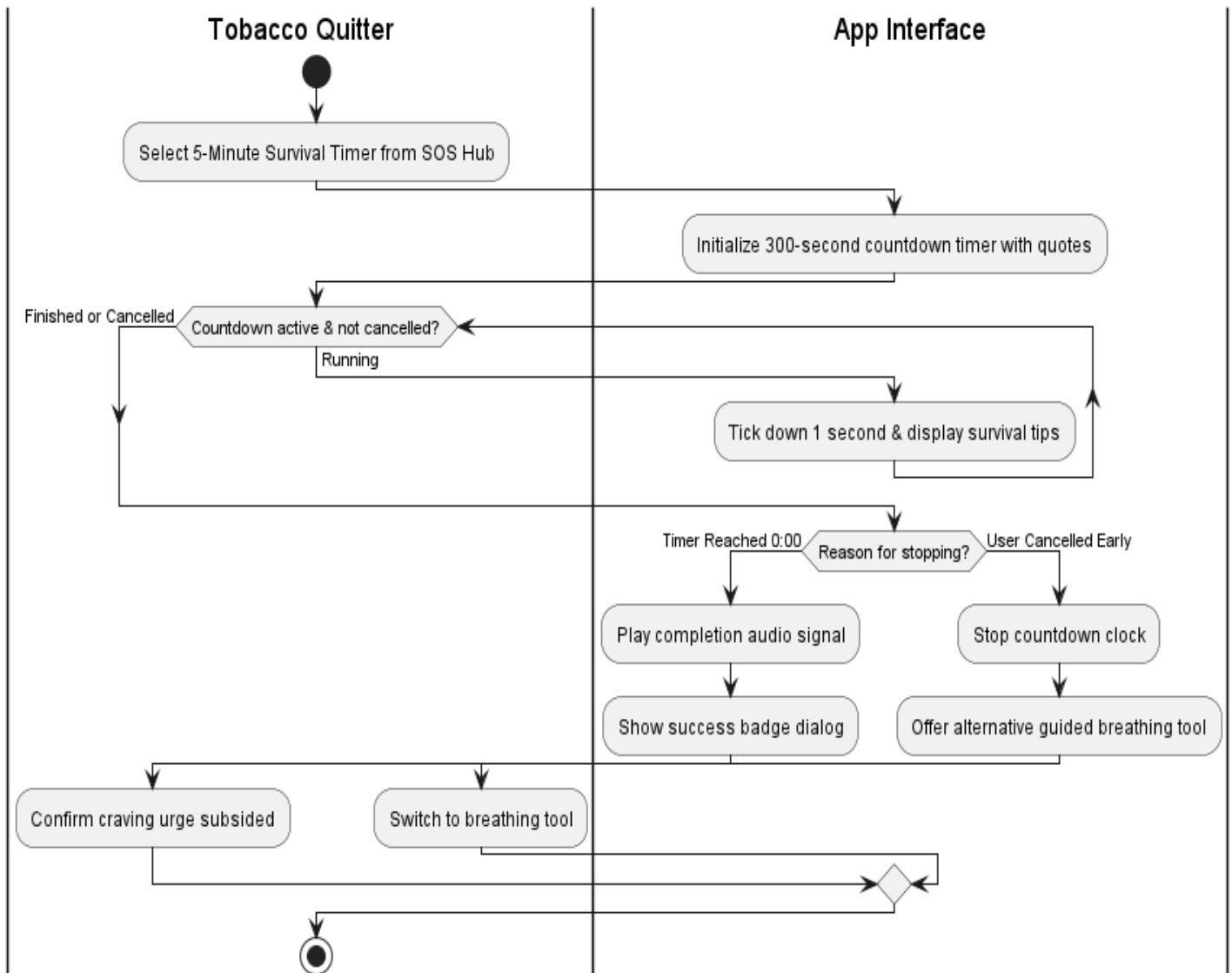


Figure 28: Swimlane Diagram for Start Survival Timer

9.8 Swimlane Diagram for Breathing Exercise

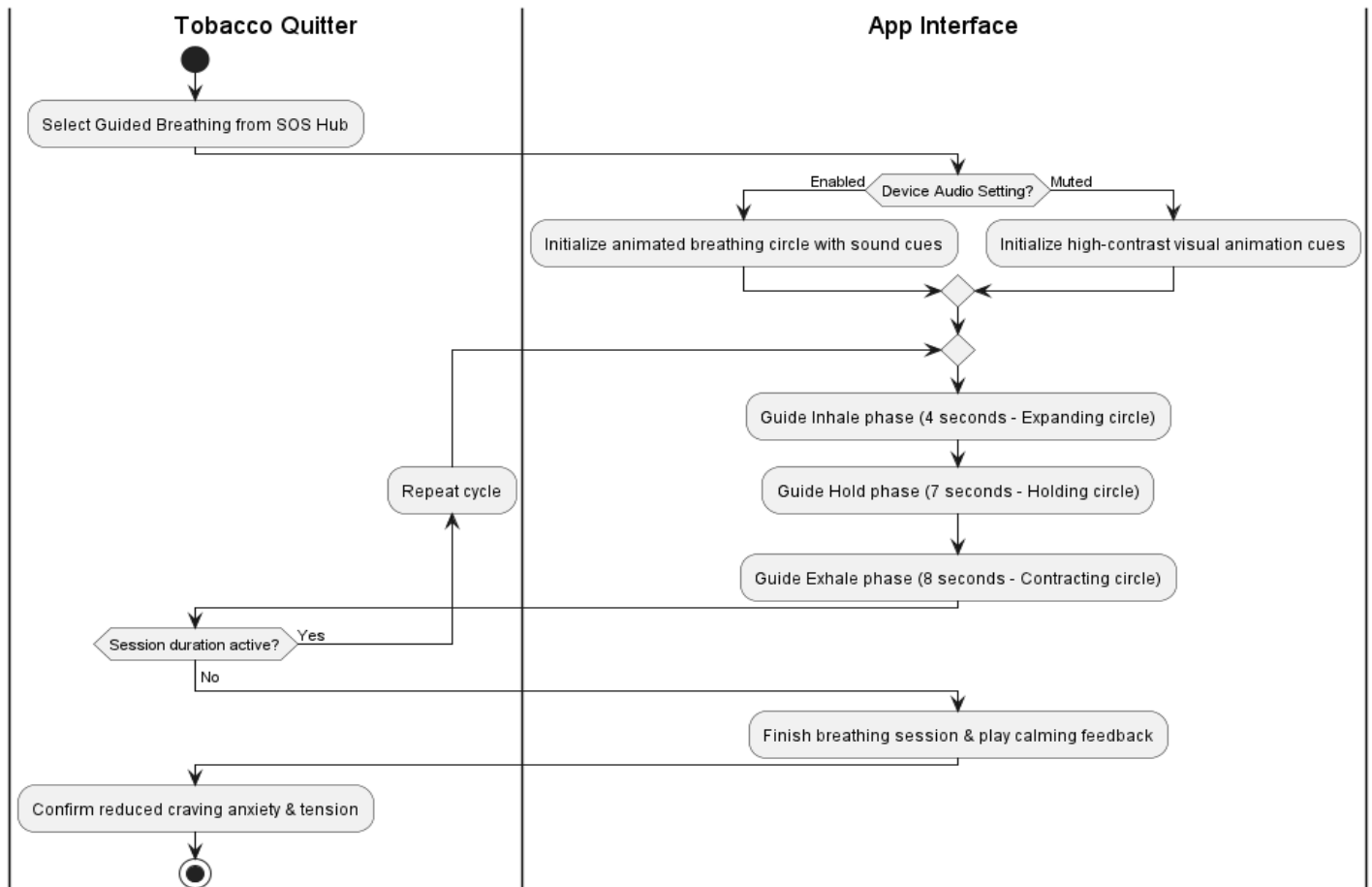


Figure 29: Swimlane Diagram for Breathing Exercise

9.9 Swimlane Diagram for Quick Distraction

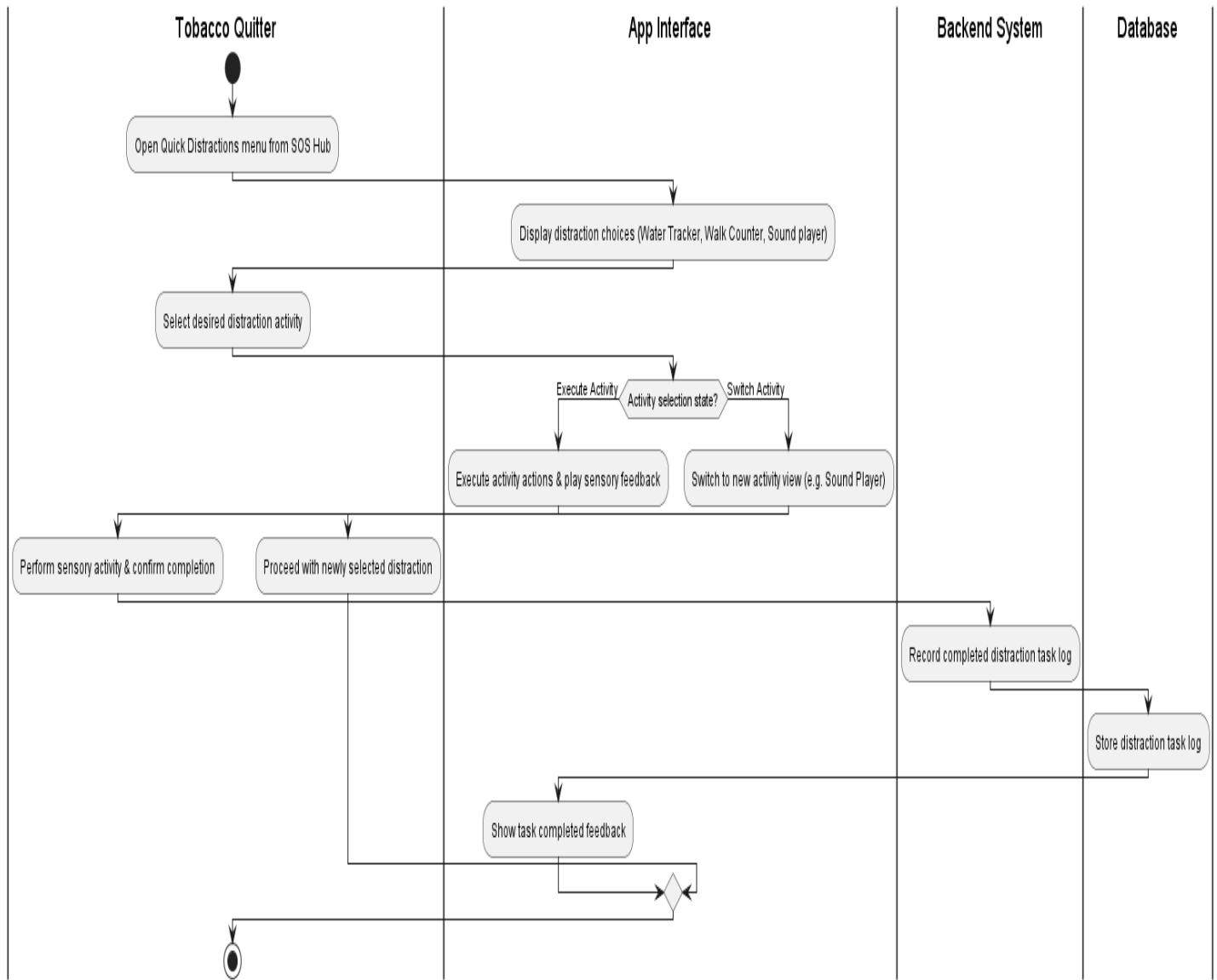


Figure 30: Swimlane Diagram for Quick Distraction

9.10 Swimlane Diagram for Access Peer Support Group

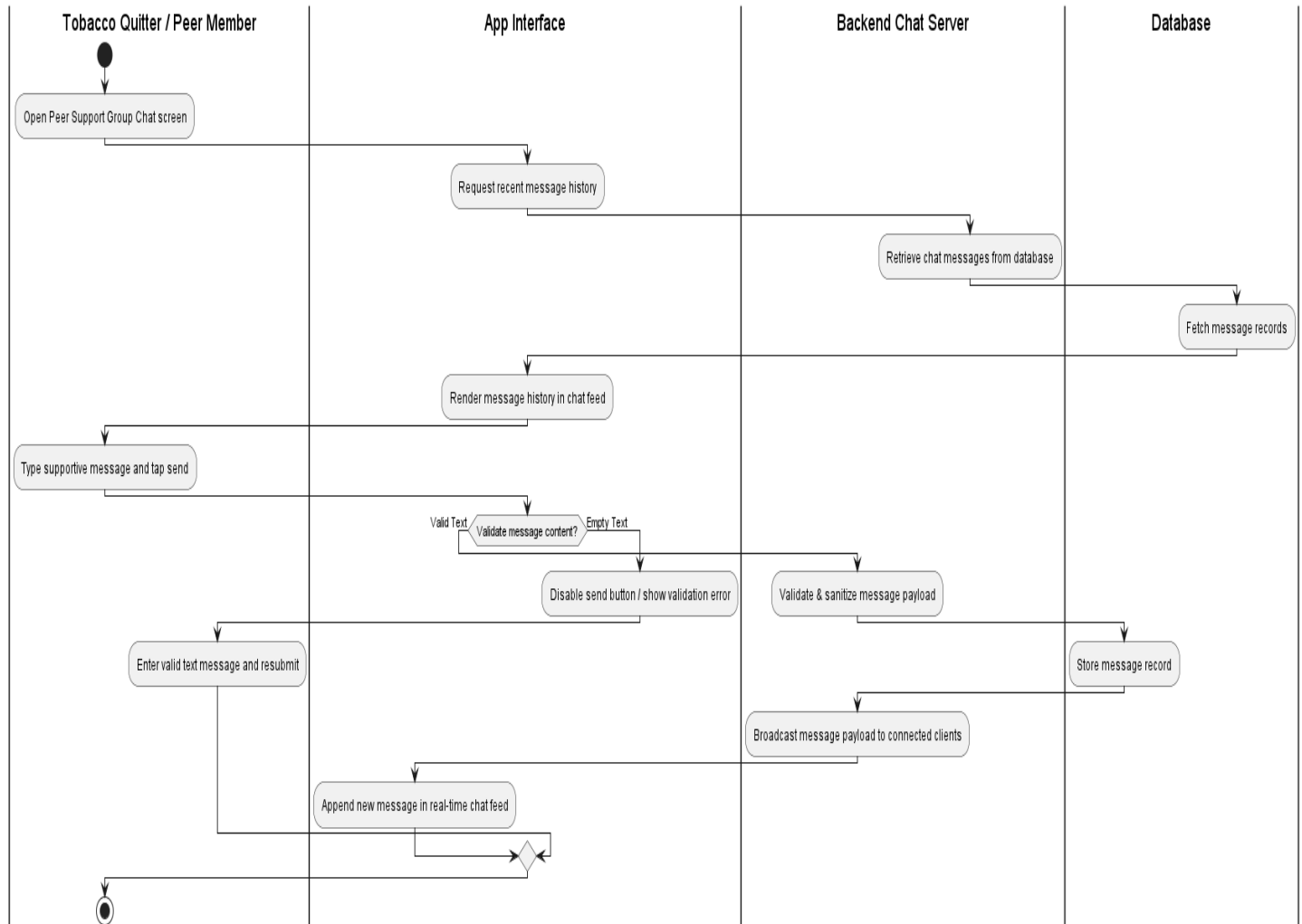


Figure 31: Swimlane Diagram for Access Peer Support Group

10. Requirement Traceability Matrix

Use Case IDs

- U1 – Manage Profile
- U2 – Set Quit Date
- U3 – View Quit Plan
- U4 – Daily Check-in
- U5 – Track Saving
- U6 – Craving Support
- U7 – Start Survival Timer
- U8 – Breathing Exercise
- U9 – Quick Distraction
- U10 – Access Peer Support Group
- U11 - View Tobacco Awareness
- U12 - View Gamification Progress
- U13 - Interact with AI Craving Companion (Groq AI)
- U14 - Receive Peer Support Group Messages
- U15 - Offline Data Caching & Auto-Synchronization
- U16 - Receive Local & Push Notifications
- U17 - Educational Content Management
- U18 – System Administration & Community Moderation

Functional Requirements ID:

- FR1 – Secure user authentication and account management.
- FR2 – Profile details updating and quit date scheduling.
- FR3 – Emergency survival timer and SOS triggering mechanism.
- FR4 – Guided breathing exercise session playback.
- FR5 – Financial savings calculation and wishlist milestone tracker.
- FR6 – Peer chatroom functionality and direct peer support messaging.
- FR7 – Push notifications and motivational alert scheduling.

FR8 – Experience points (Streak Badges) and gamification logic calculations.

FR9 – Offline Data Caching & Auto-Synchronization: The system shall locally cache user entries in Hive offline storage, auto-syncing with the PostgreSQL backend when internet connectivity is restored.

RTM

Req ID	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	U11	U12	U13	U14	U15	U16	U17	U18
FR1	✓																	
FR2		✓	✓															
FR3						✓	✓											
FR4								✓										
FR5					✓													
FR6										✓	✓							✓
FR7														✓				
FR8													✓					
FR9																✓	✓	✓

Figure 32: Requirement Traceability Matrix

11. Database Design

11.1 Entity Relationship Diagram

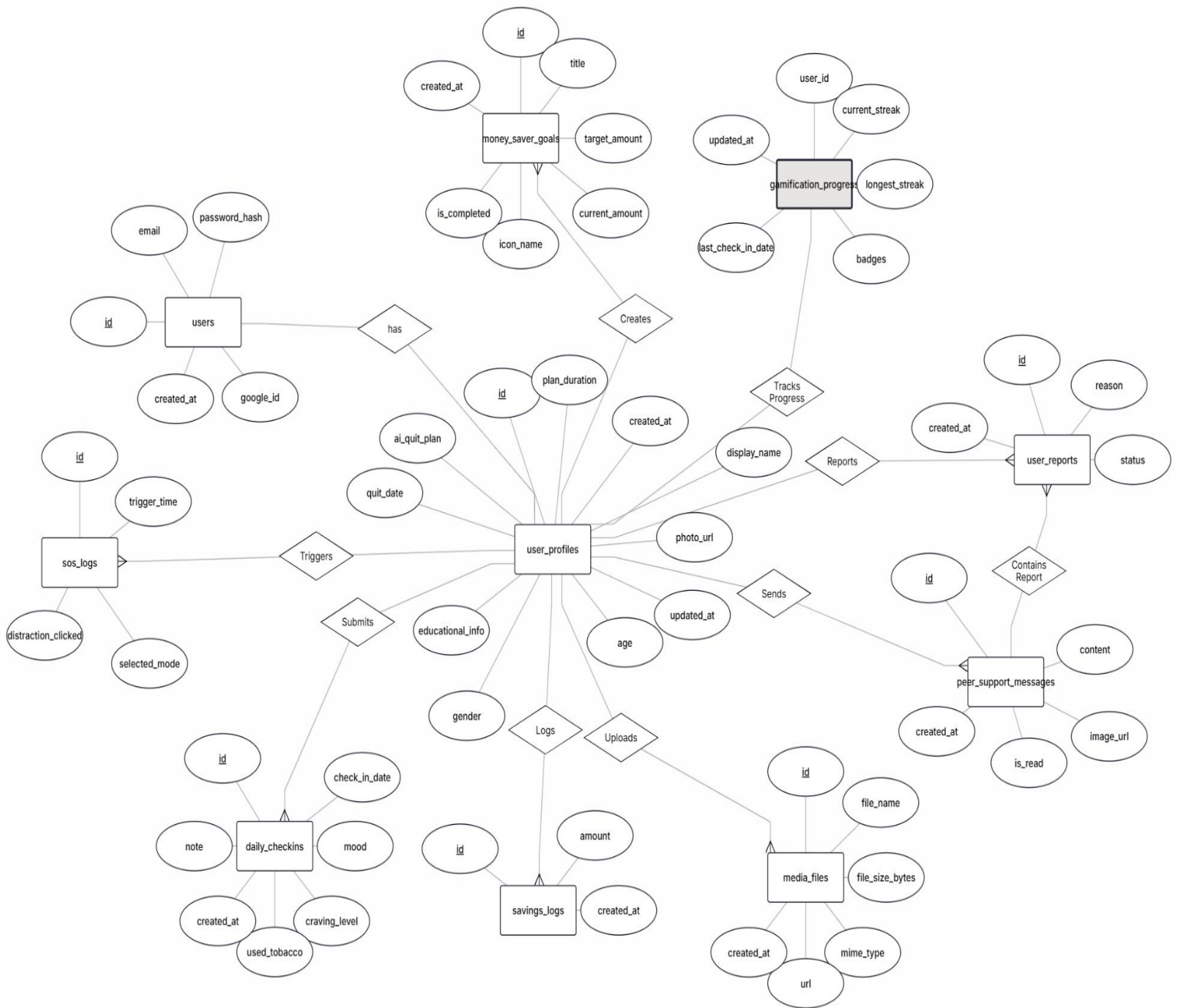


Figure 33: Entity Relationship Diagram

11.2 ER Diagram to Relational Schema:

1. users(id[PK], email, password_hash, google_id, created_at)
2. user_profiles(id[PK, FK -> users.id], display_name, photo_url, age, gender, educational_info, plan_duration, quit_date, ai_quit_plan, created_at, updated_at)
3. daily_checkins(id[PK], user_id[FK -> user_profiles.id], check_in_date, mood, craving_level, used_tobacco, note, created_at)
4. gamification_progress(user_id[PK, FK -> user_profiles.id], longest_streak, current_streak, badges[TEXT[]], last_check_in_date, updated_at)
5. money_saver_goals(id[PK], user_id[FK -> user_profiles.id], title, target_amount, current_amount, icon_name, is_completed, created_at)
6. savings_logs(id[PK], user_id[FK -> user_profiles.id], amount, created_at)
7. peer_support_messages(id[PK], sender_id[FK -> user_profiles.id], content, image_url, is_read, created_at)
8. media_files(id[PK], user_id[FK -> user_profiles.id], file_name, file_size_bytes, mime_type, url, created_at)
9. sos_logs(id[PK], user_id[FK -> user_profiles.id], trigger_time, selected_mode, distraction_clicked)
10. user_reports(id[PK], reporter_id[FK -> user_profiles.id], reported_user_id[FK], message_id[FK], reason, status, created_at)

12 Conclusion

The Software Requirements Specification (SRS) for the QuitMate - Tobacco Awareness mobile application provides a detailed roadmap for the development and implementation of a comprehensive tobacco cessation platform. By identifying the specific needs of tobacco quitters, peer group members, and medical advisors, this document ensures that the system is built to address the physical, psychological, and financial challenges of overcoming addiction.

The integration of features such as personalized quit plans, emergency craving support tools, financial tracking, and gamification creates a supportive environment for long-term behavior change. Furthermore, by maintaining high standards for data security, system performance, and user privacy, the application is positioned to be a reliable tool in public health management. This specification serves as the foundation for the development phase, ensuring that all stakeholders have

a shared understanding of the system's goals to successfully help users achieve a tobacco-free lifestyle.

13 Reference

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14 Appendix

Term	Meaning
Tobacco Awareness	The tobacco quitting mobile application.
Tobacco Quitter	A person trying to quit tobacco products.
Quit Date	The selected date to stop using tobacco.
Craving	A strong psychological or physical urge to use tobacco.
Daily Check-in	A daily feature where users record their tobacco usage status and mood.
SOS Button	An emergency support feature used during intense cravings.
Survival Timer	A 5-minute countdown timer used during cravings to distract the user.
Breathing Exercise	The 4-7-8 guided breathing method to reduce stress and anxiety.
Money Saver Tracker	A feature that calculates the money saved by avoiding tobacco.
Wishlist	An item or goal the tobacco quitter wants to achieve using their saved money.
Streak Badges	Experience points earned by completing daily activities and milestones.
Peer Group Member	A support person who provides emotional guidance to tobacco quitters.
Moderator	A person who monitors group discussions to ensure safety and policy compliance.
Medical Advisor	A professional who verifies quitting guidelines and awareness content.
System Administrator	A person responsible for managing and maintaining the entire application system.
Stakeholder	Any individual or organization involved in or affected by the project.
SRS	Software Requirements Specification – the documentation of system requirements.